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(54) SEMICONDUCTOR DEVICE STRUCTURE AND METHOD FOR FORMING THE SAME

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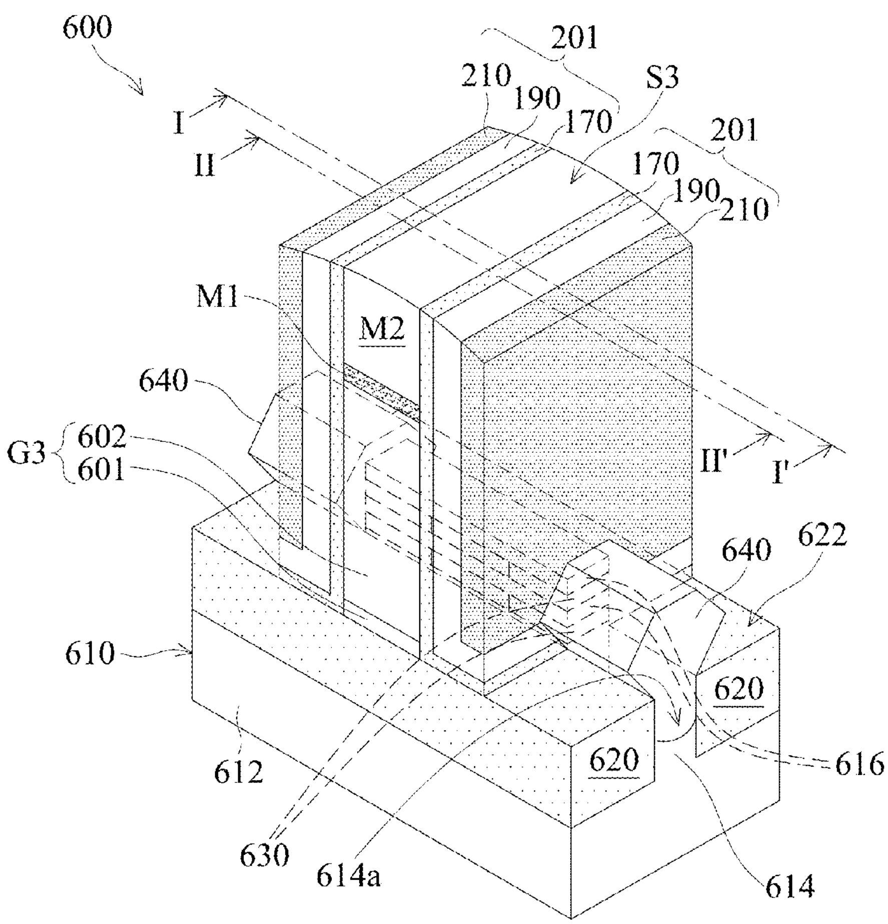
(57) **ABSTRACT**

A method for forming a semiconductor device structure is provided. The method includes forming a gate stack over a substrate. The method includes forming a spacer structure over a sidewall of the gate stack. The method includes forming a source/drain structure in and over the substrate, wherein a portion of the spacer structure is between the source/drain structure and the gate stack. The method includes partially removing the outer layer, wherein a first lower portion of the outer layer remains between the source/drain structure and the gate stack. The method includes partially removing the middle layer, wherein a second lower portion of the middle layer remains between the source/drain structure and the gate stack.



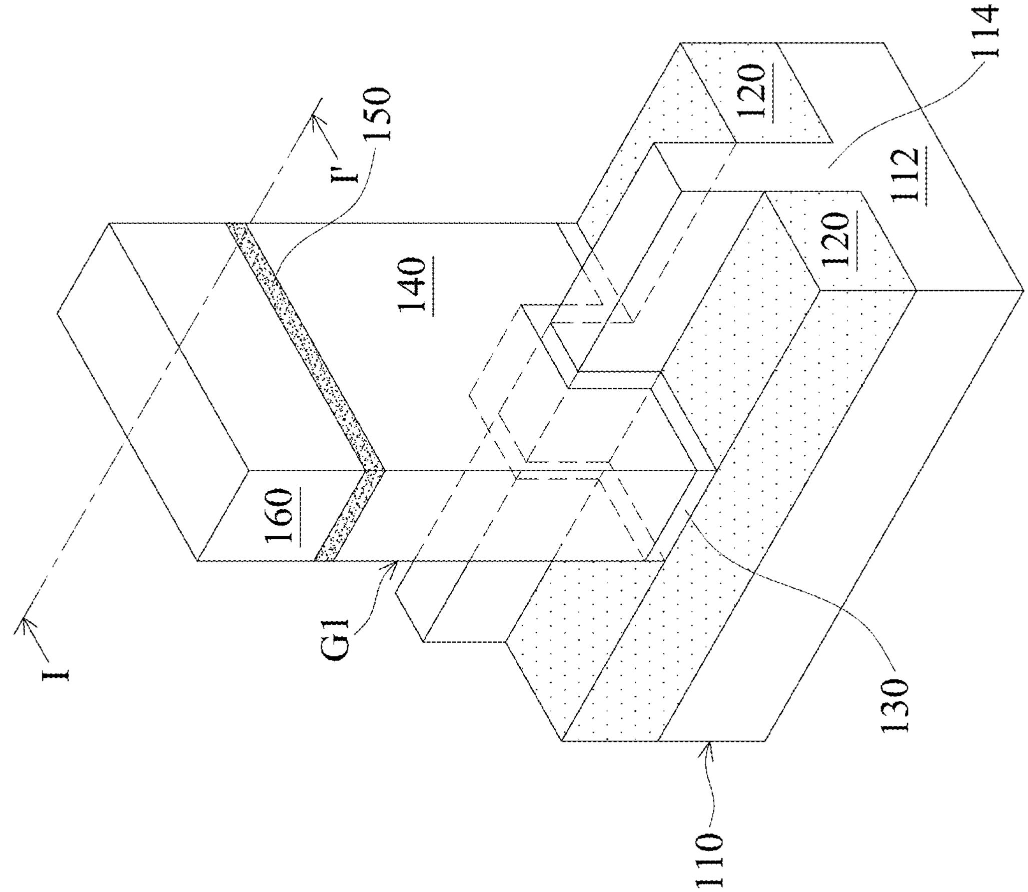

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FIG. 1

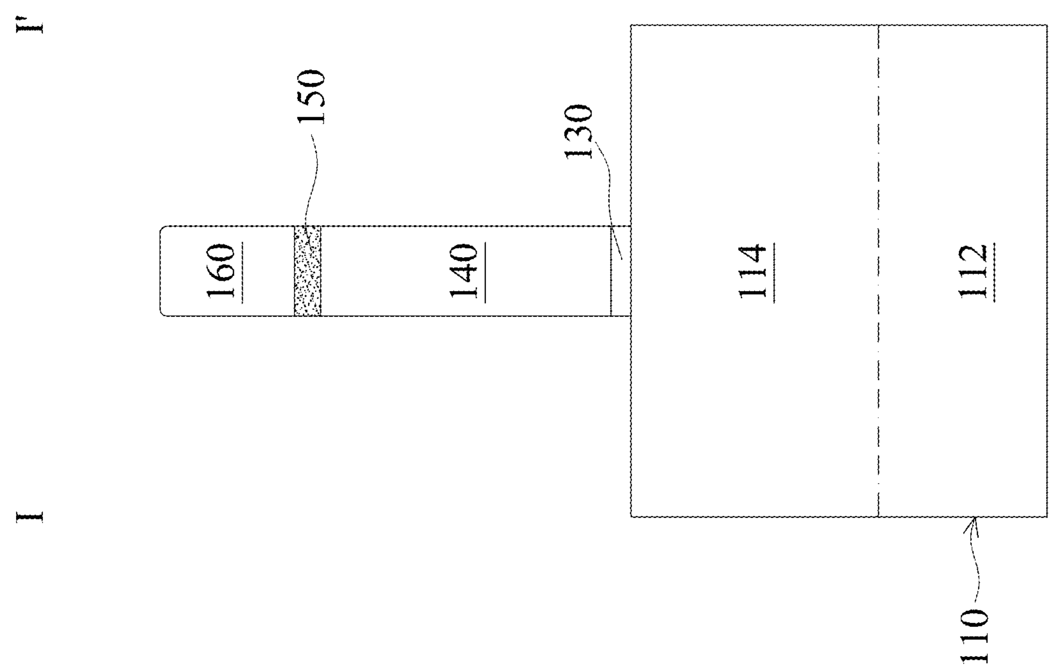
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FIG. 2A

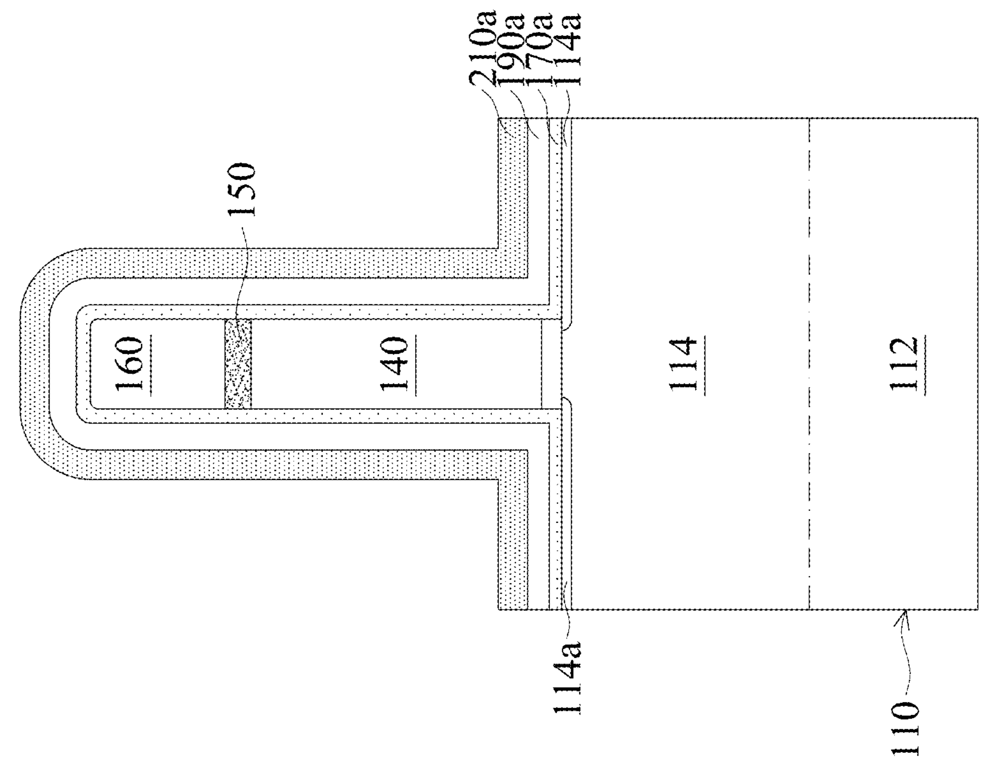
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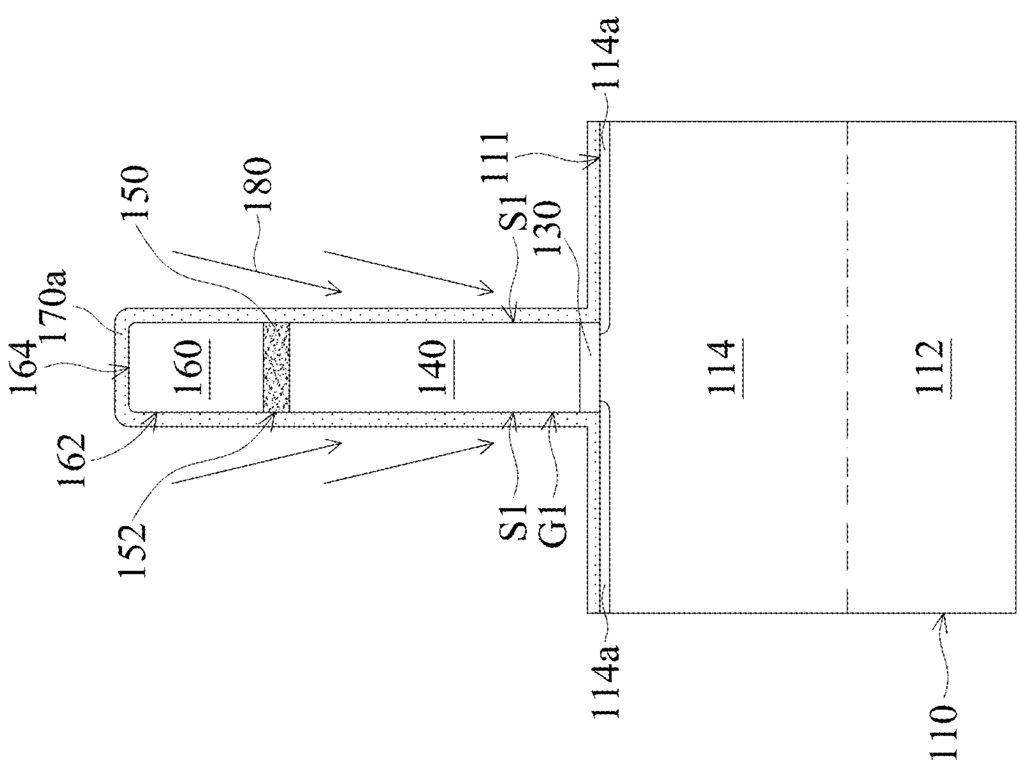
FIG. 2C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S1
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114
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114a
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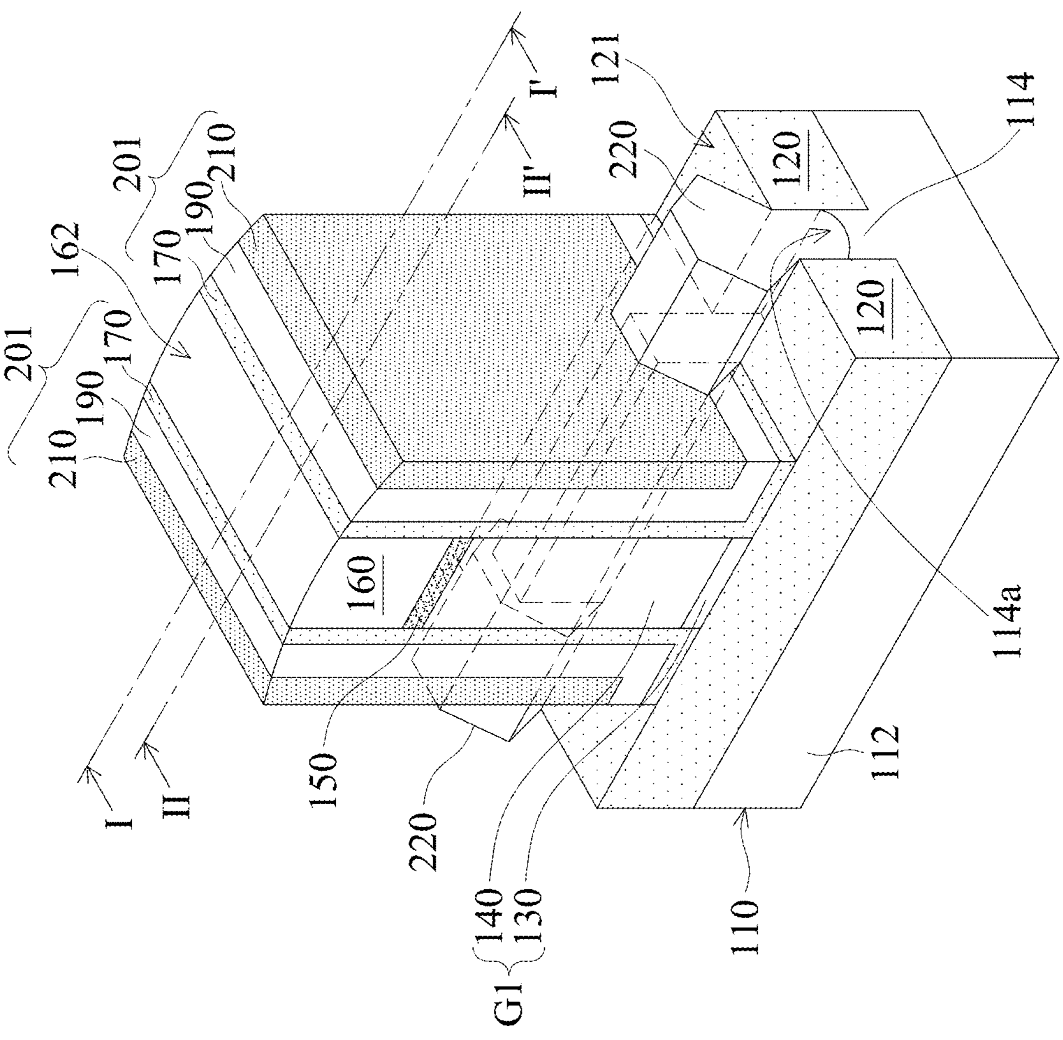
FIG. 2B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II'
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114a
112
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G1
140
130

FIG. 3

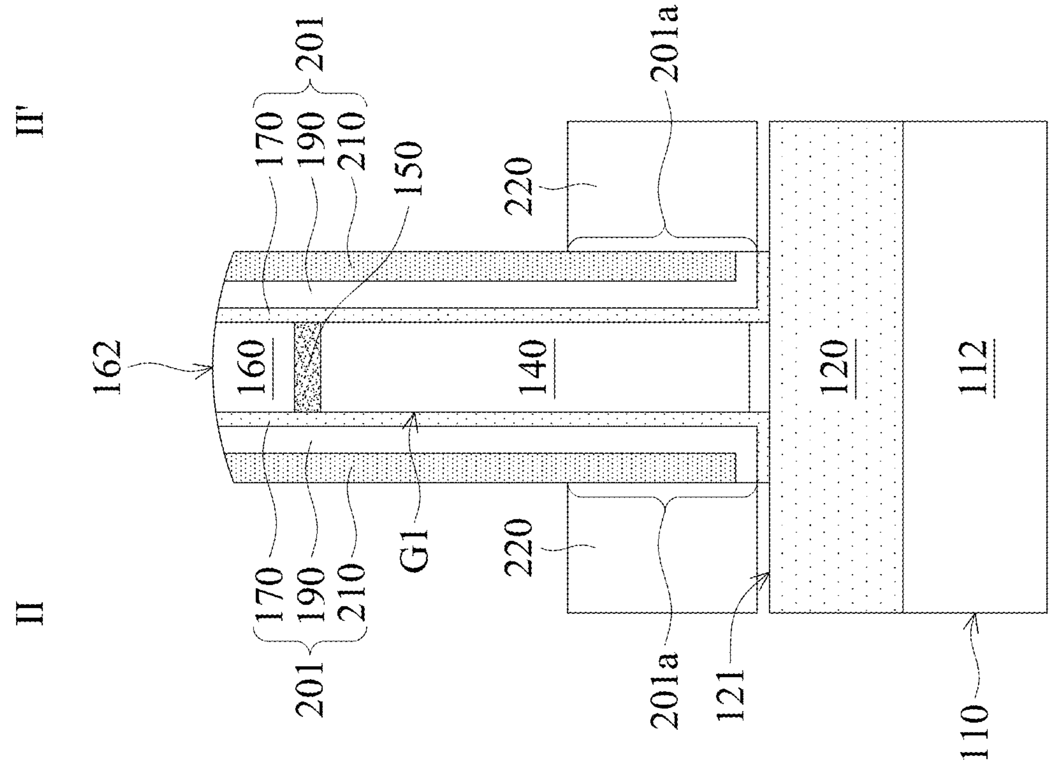

FIG. 4A-2

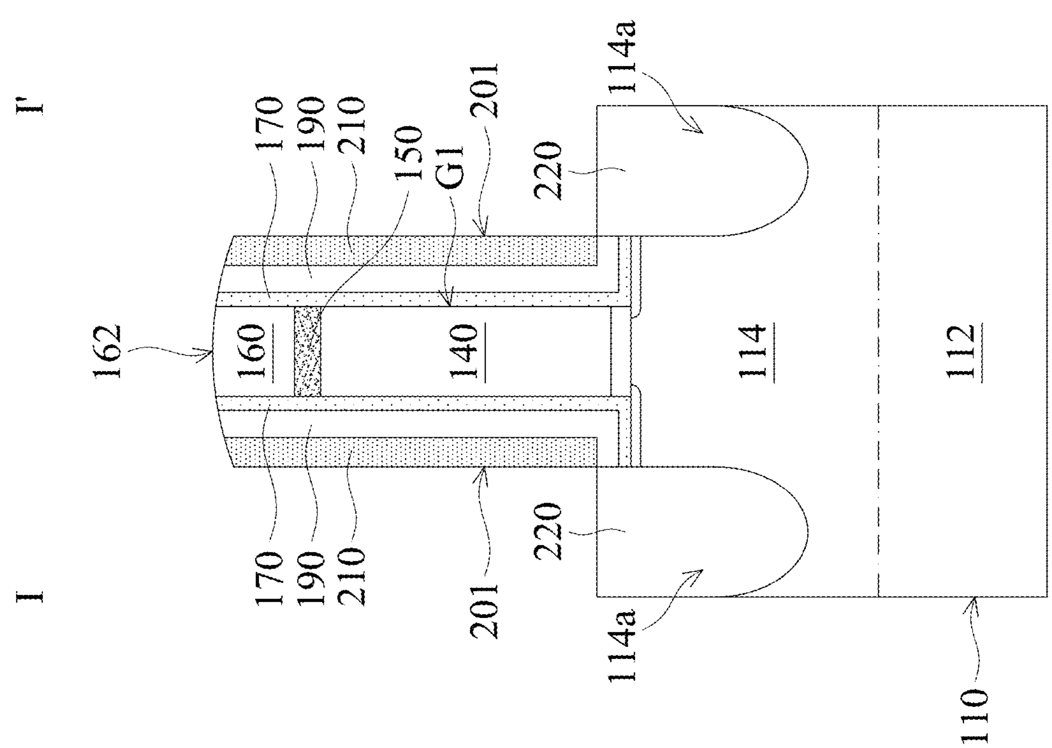

FIG. 4A-1

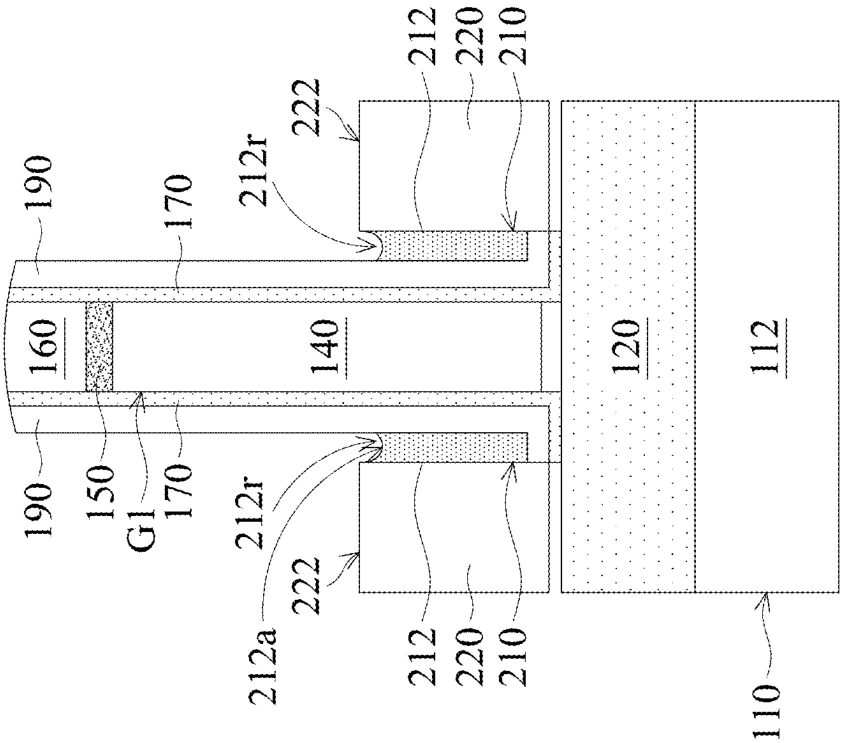

FIG. 4B-2

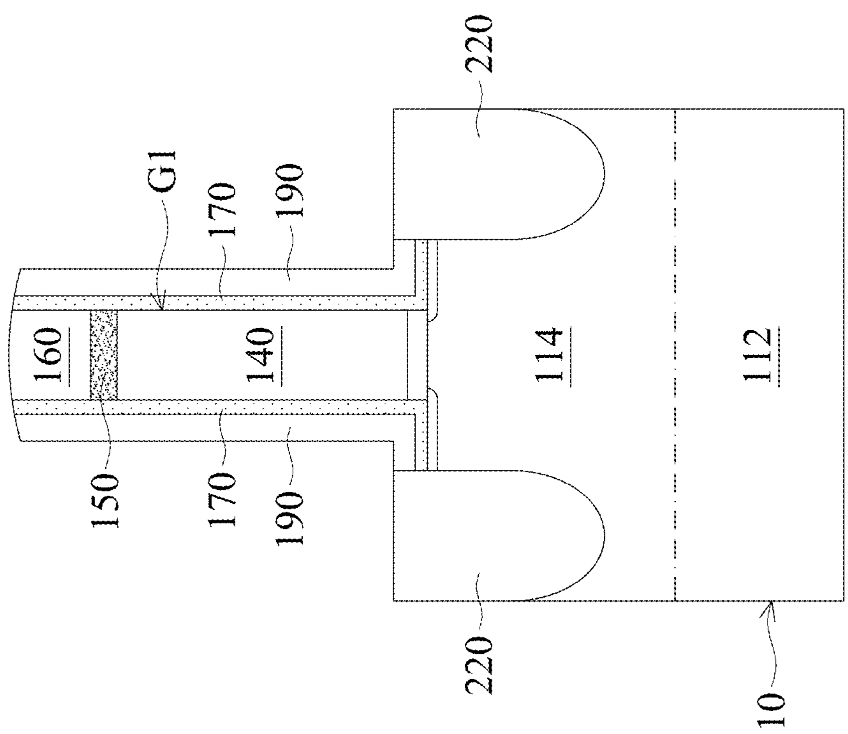

FIG. 4B-1

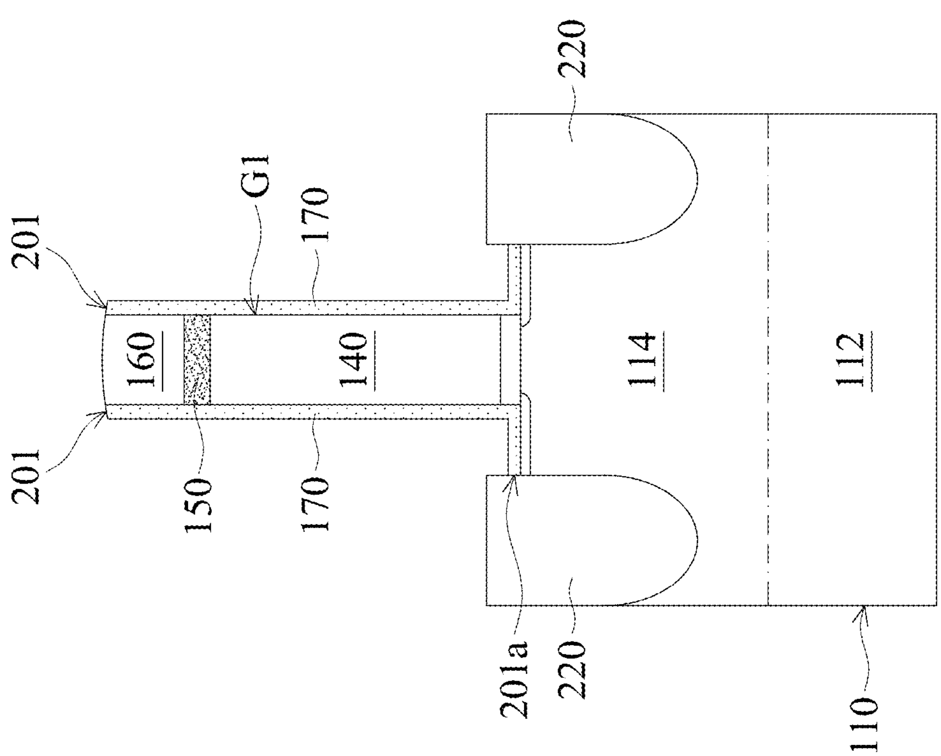

FIG. 4C-1

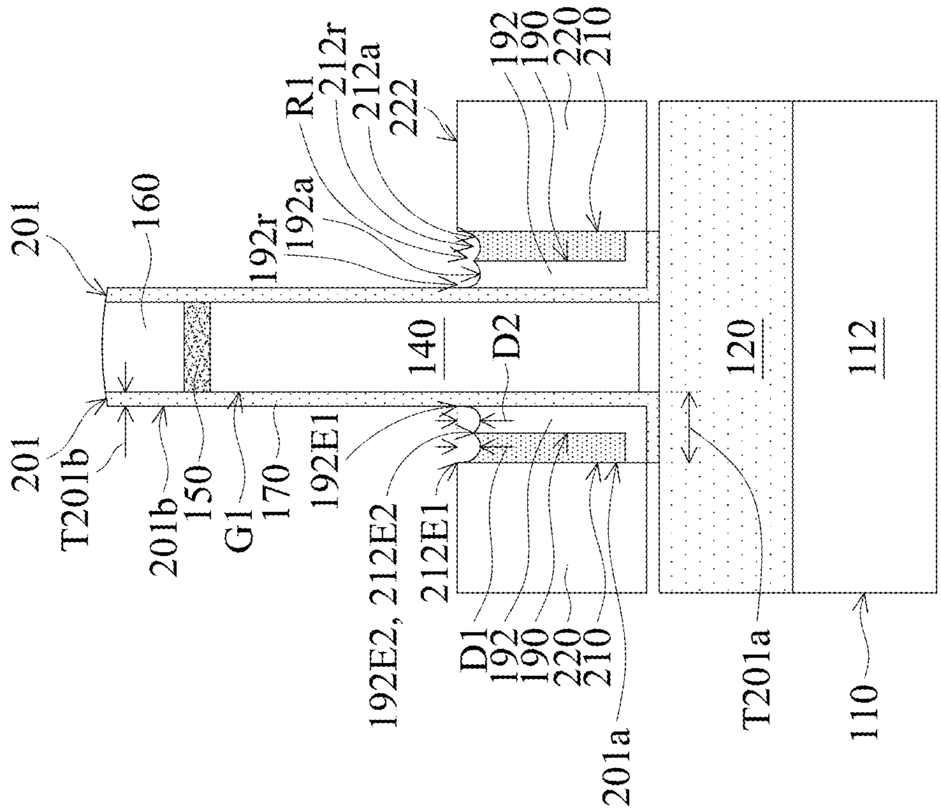

FIG. 4C-2

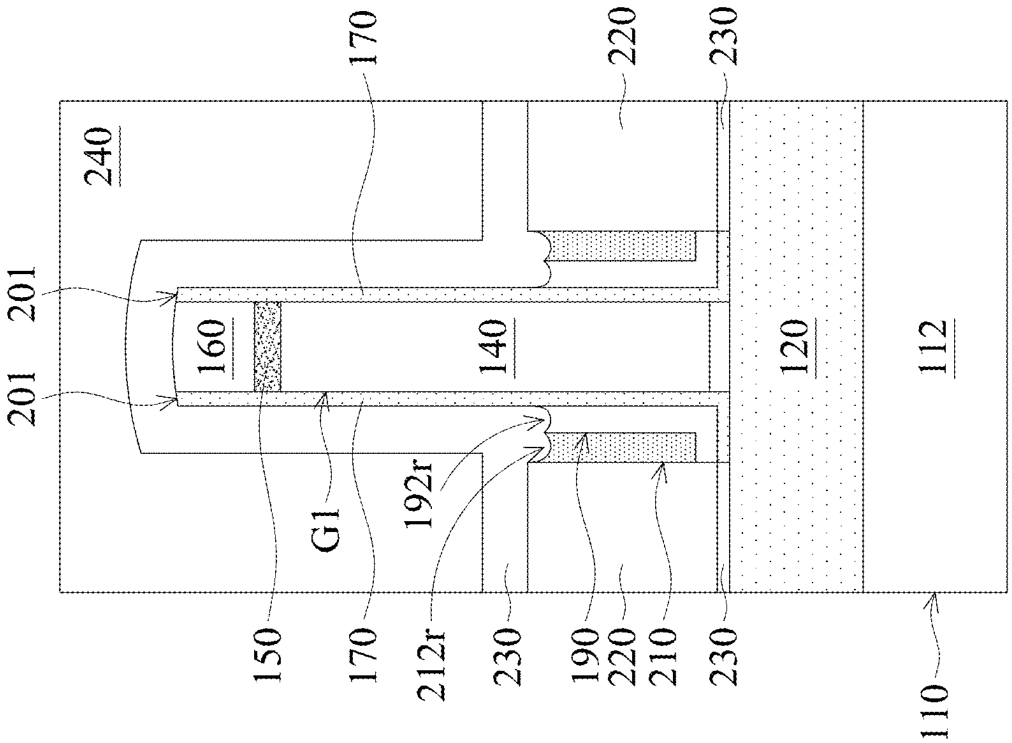

FIG. 4D-2

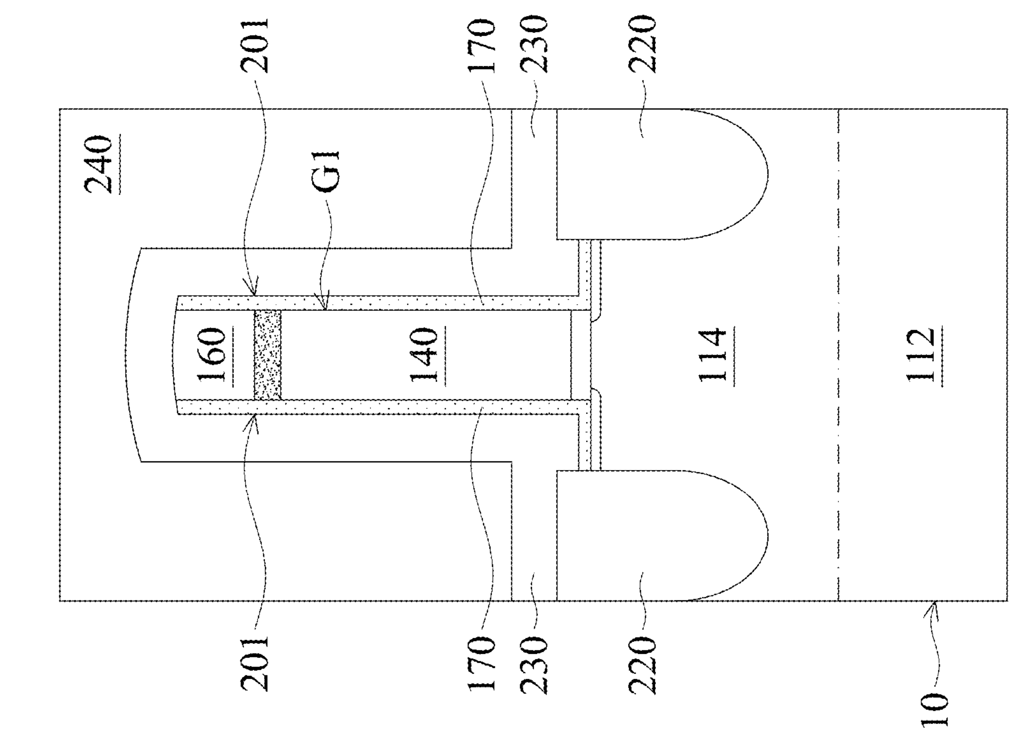

FIG. 4D-1

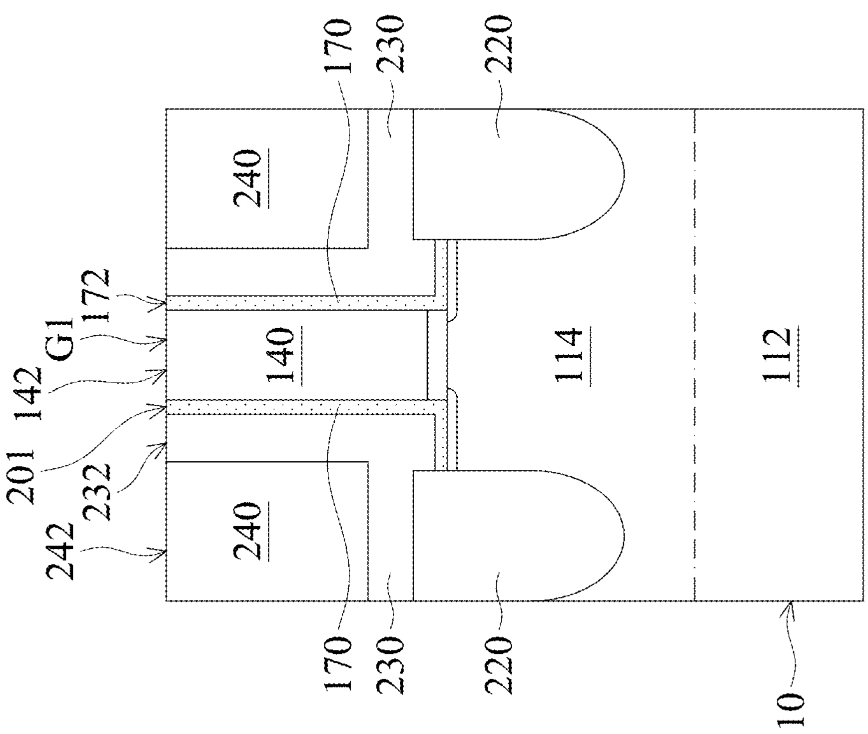

FIG. 4E-1

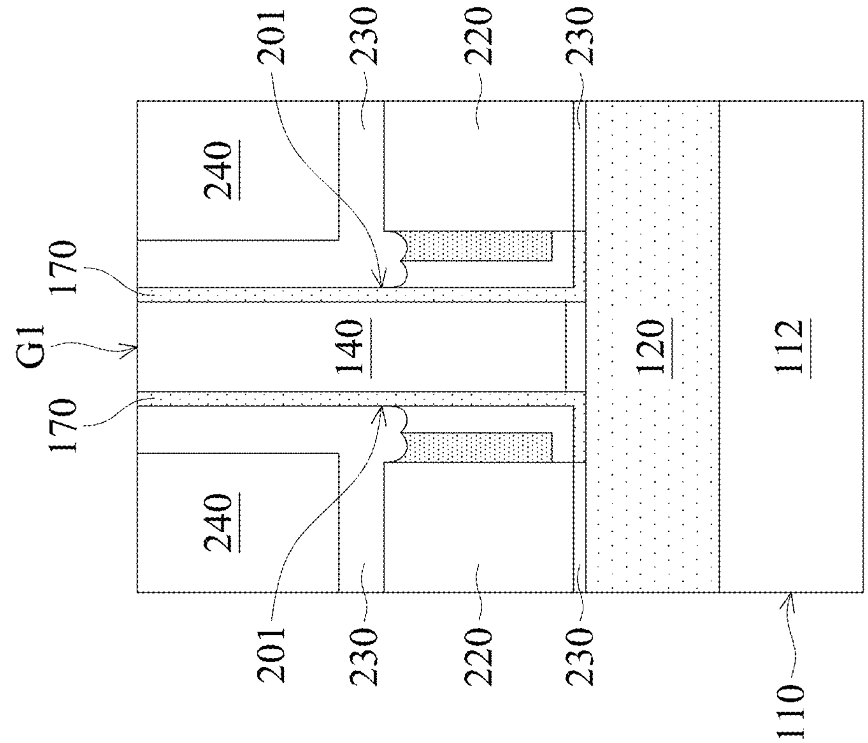

FIG. 4E-2

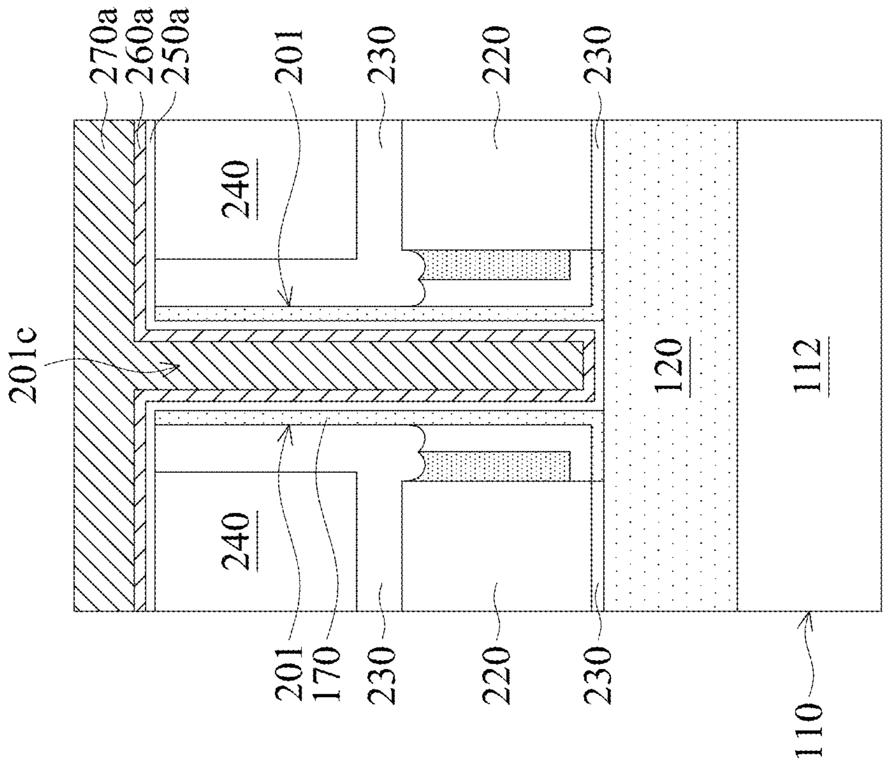

FIG. 4F-2

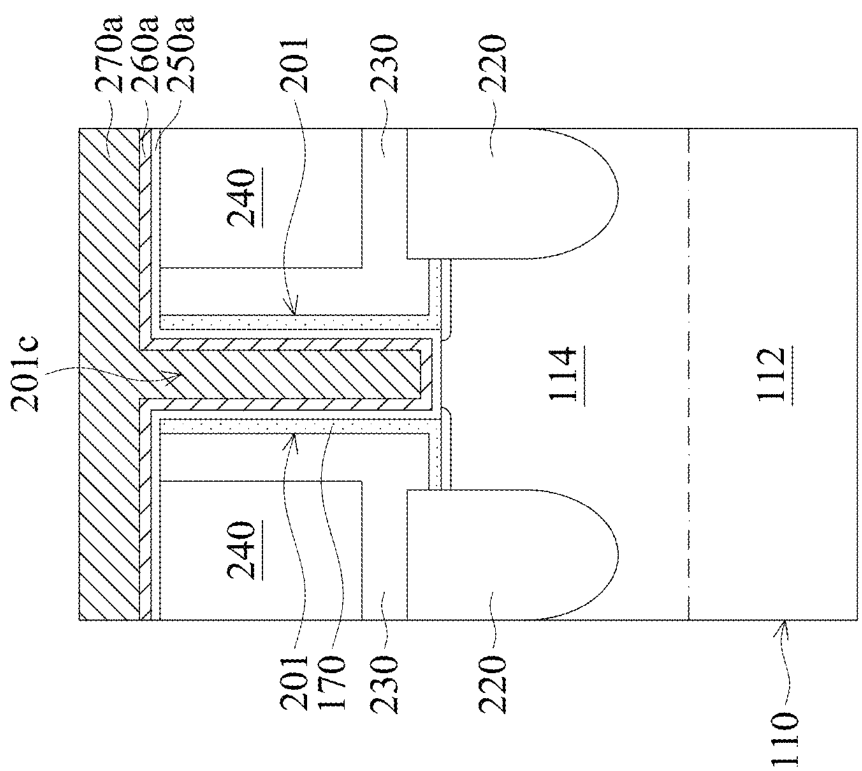

FIG. 4F-1

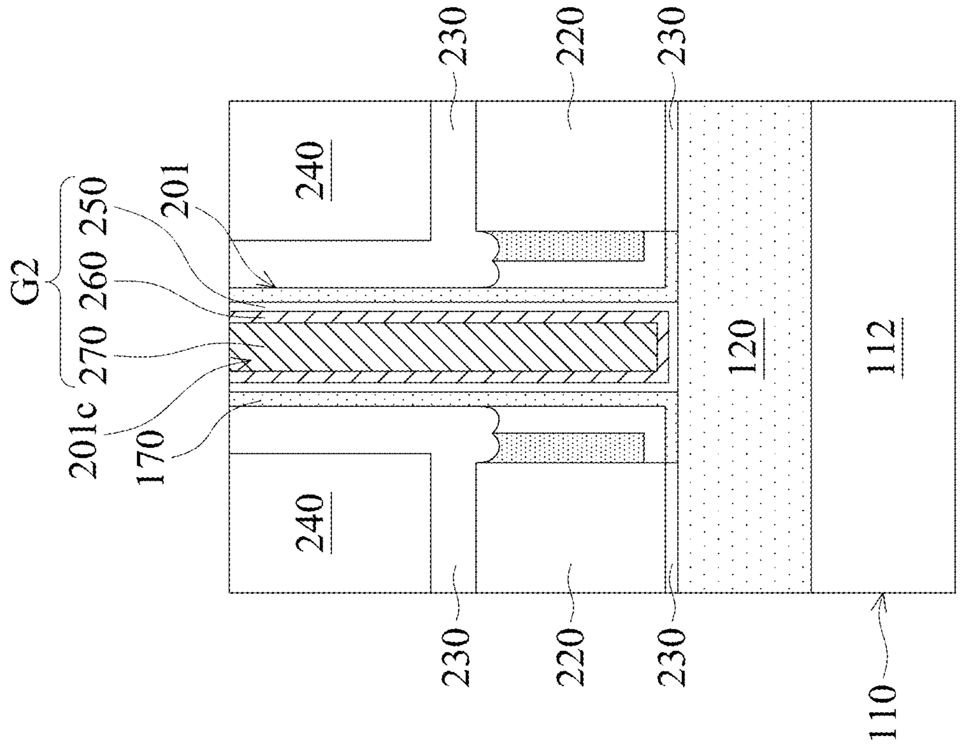

FIG. 4G-2

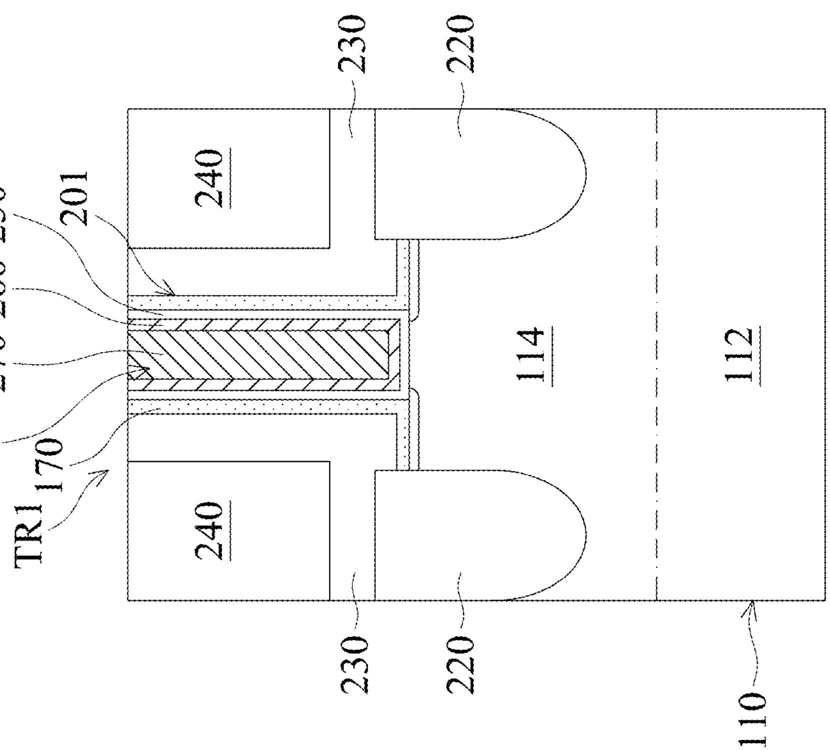

FIG. 4G-1

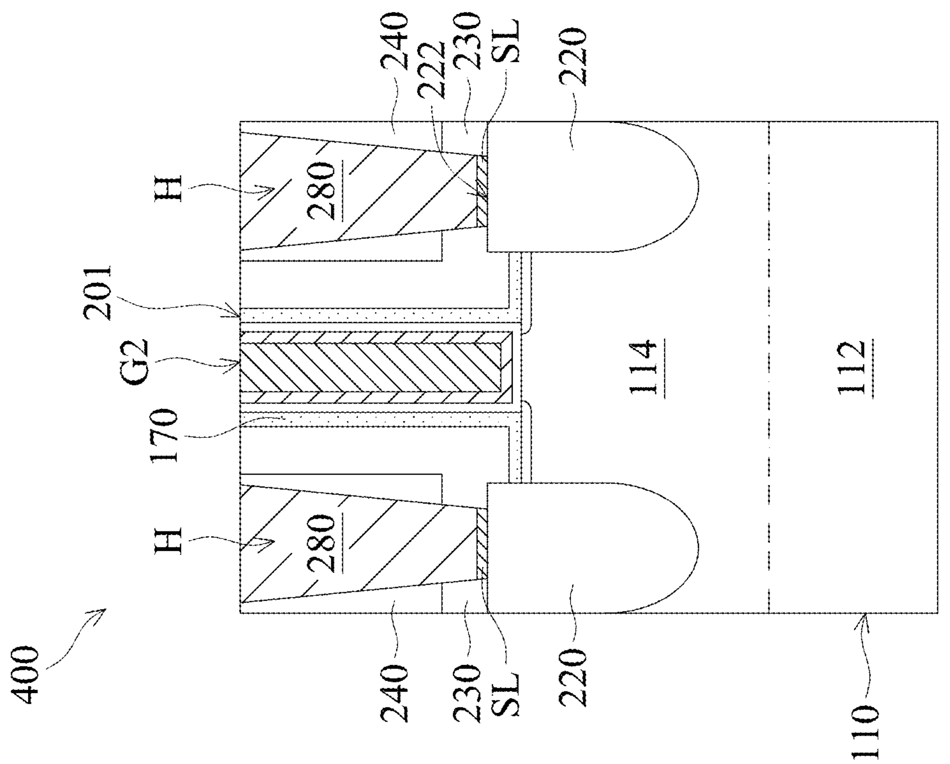

FIG. 4H-1

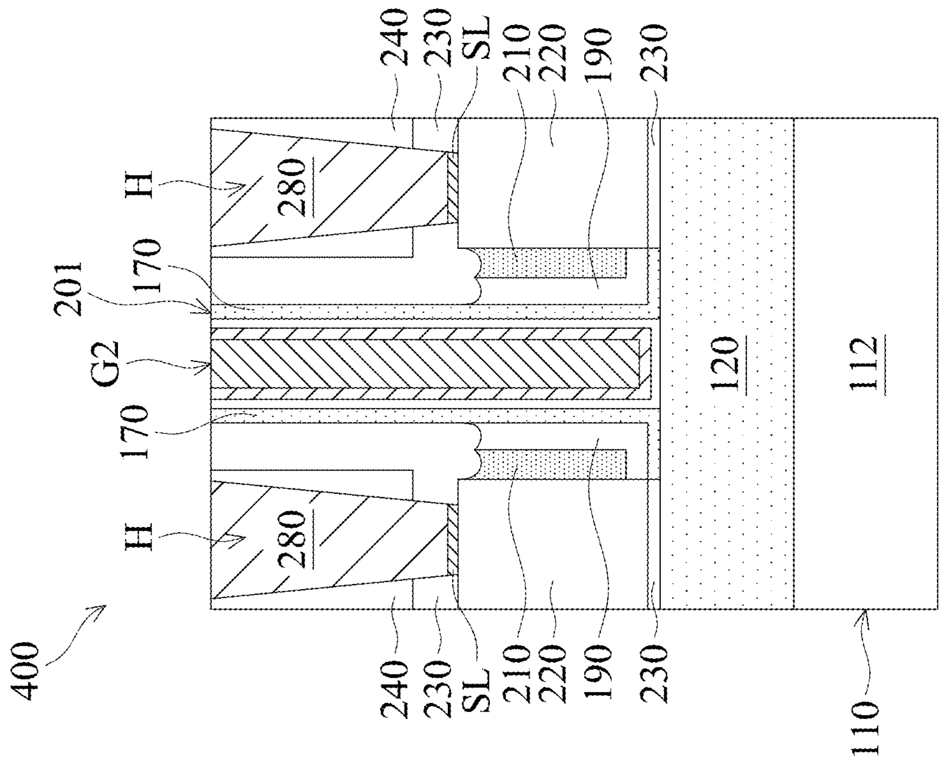

FIG. 4H-2

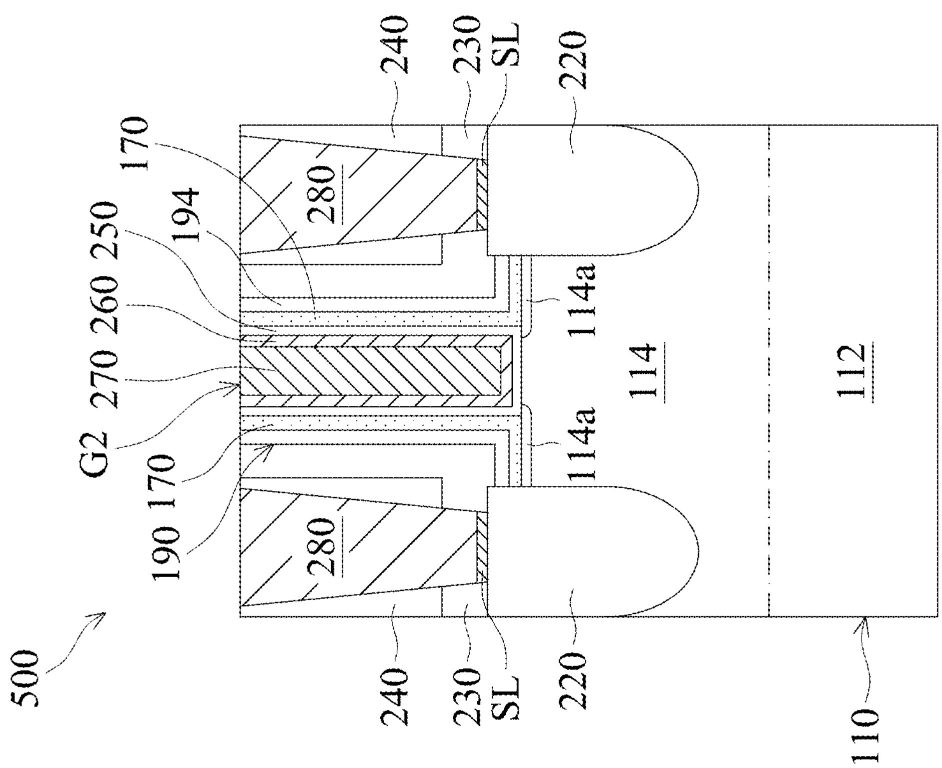
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FIG. 5A

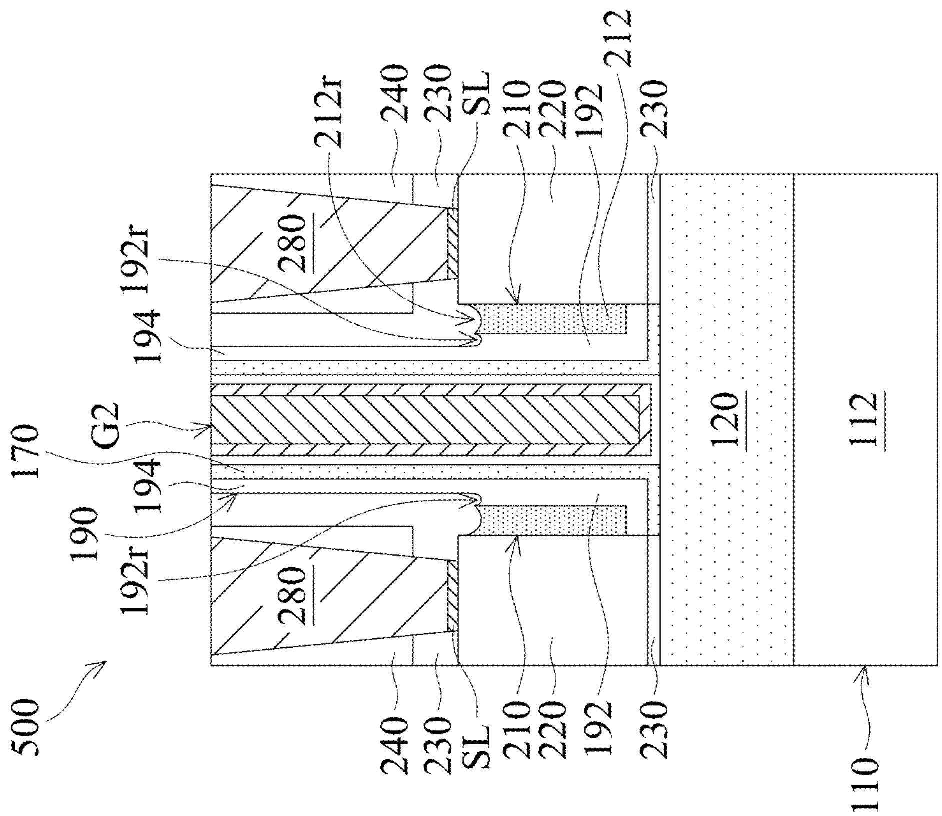
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FIG. 5B

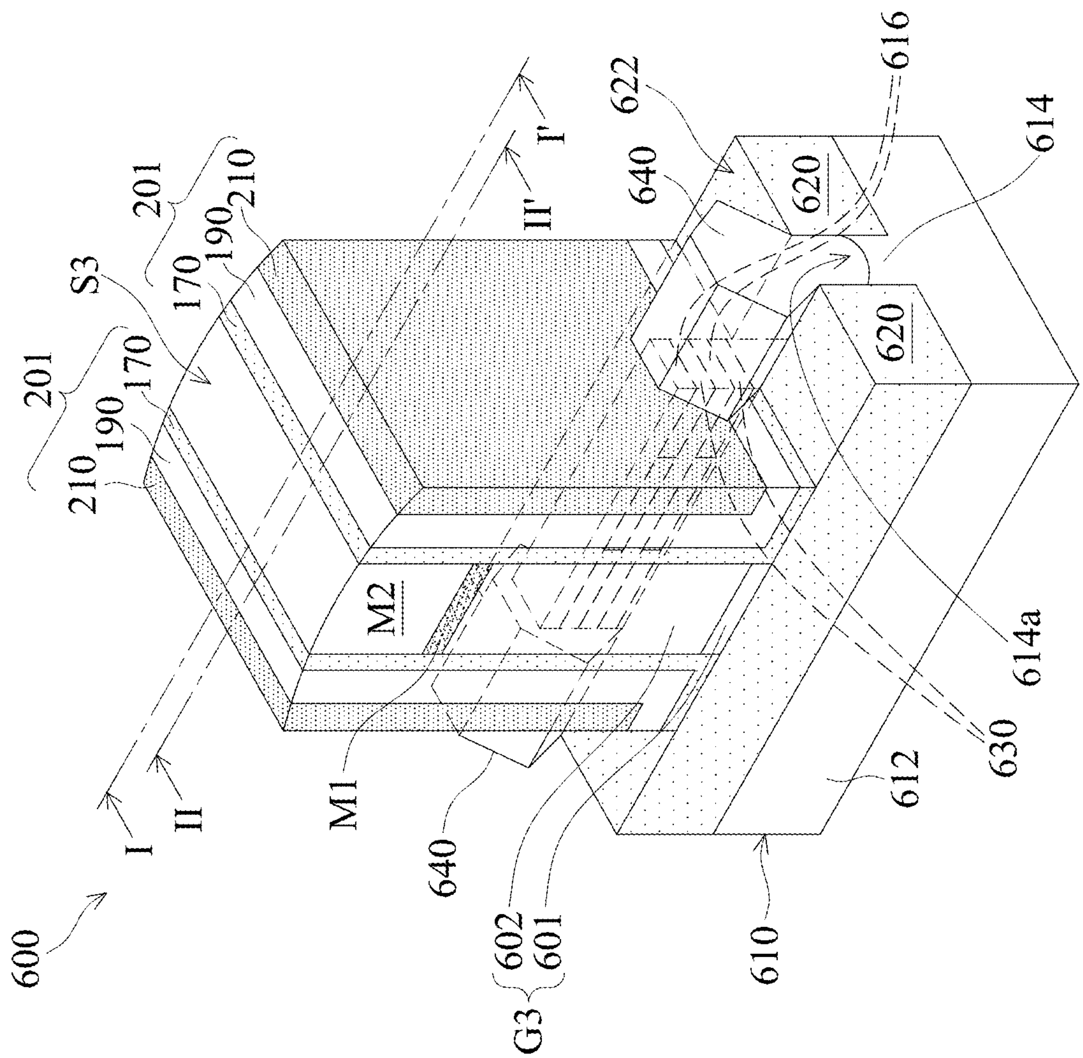
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FIG. 6

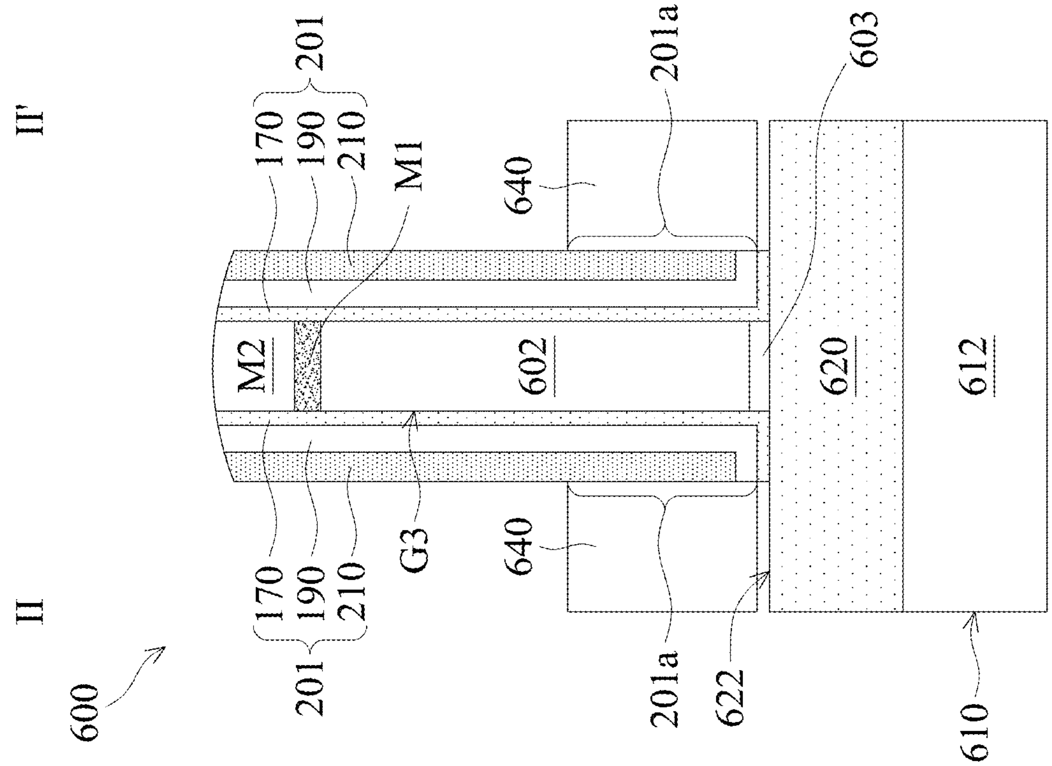

FIG. 7A-2

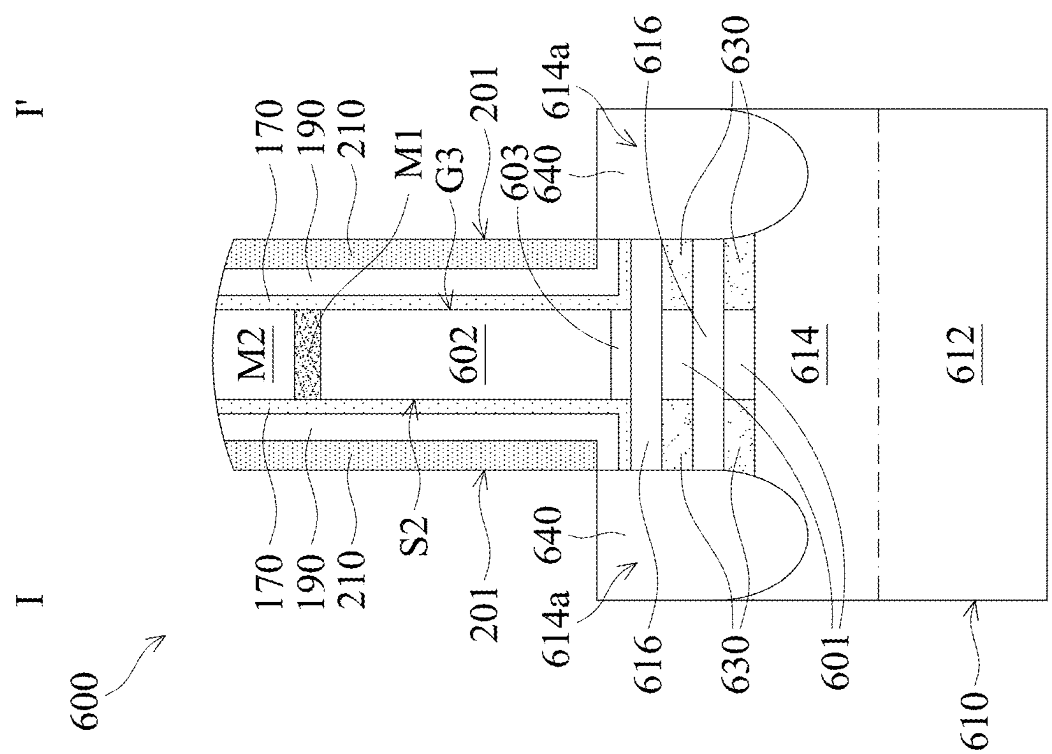

FIG. 7A-1

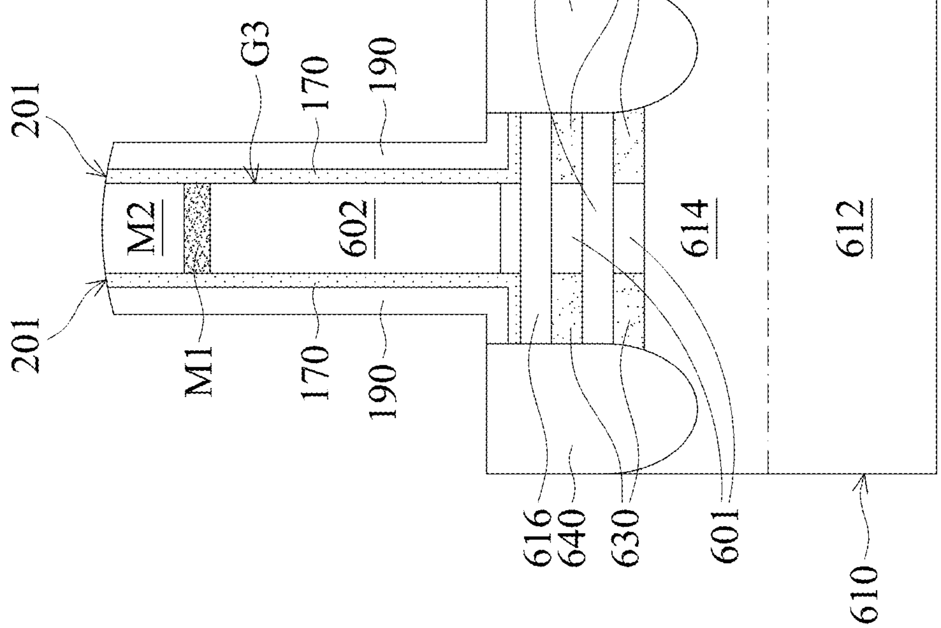

FIG. 7B-1

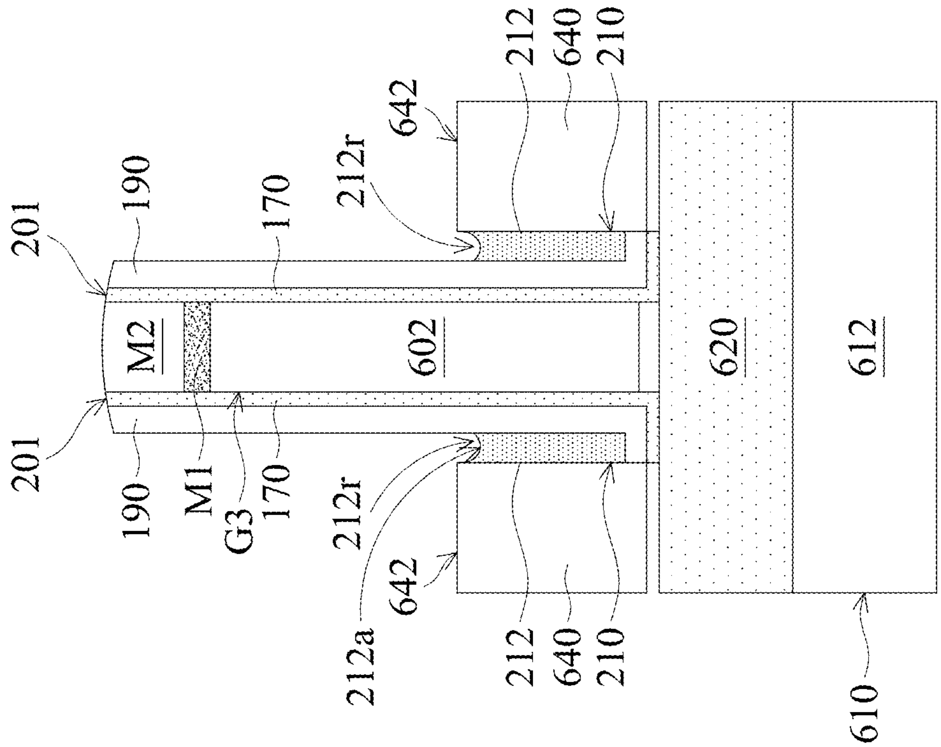

FIG. 7B-2

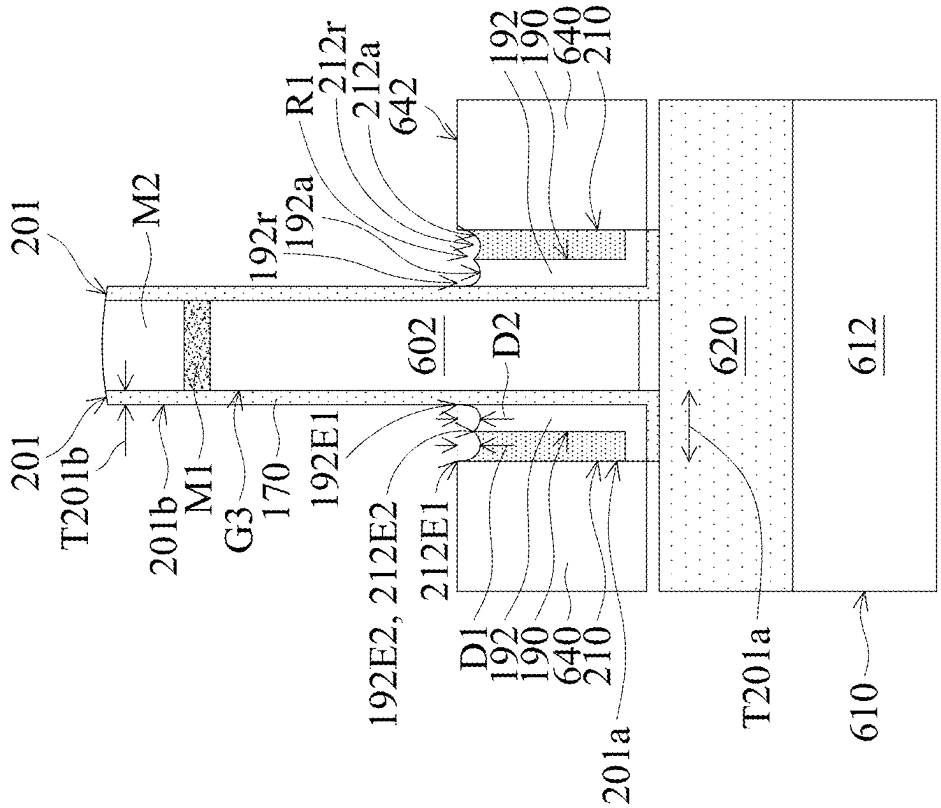

FIG. 7C-2

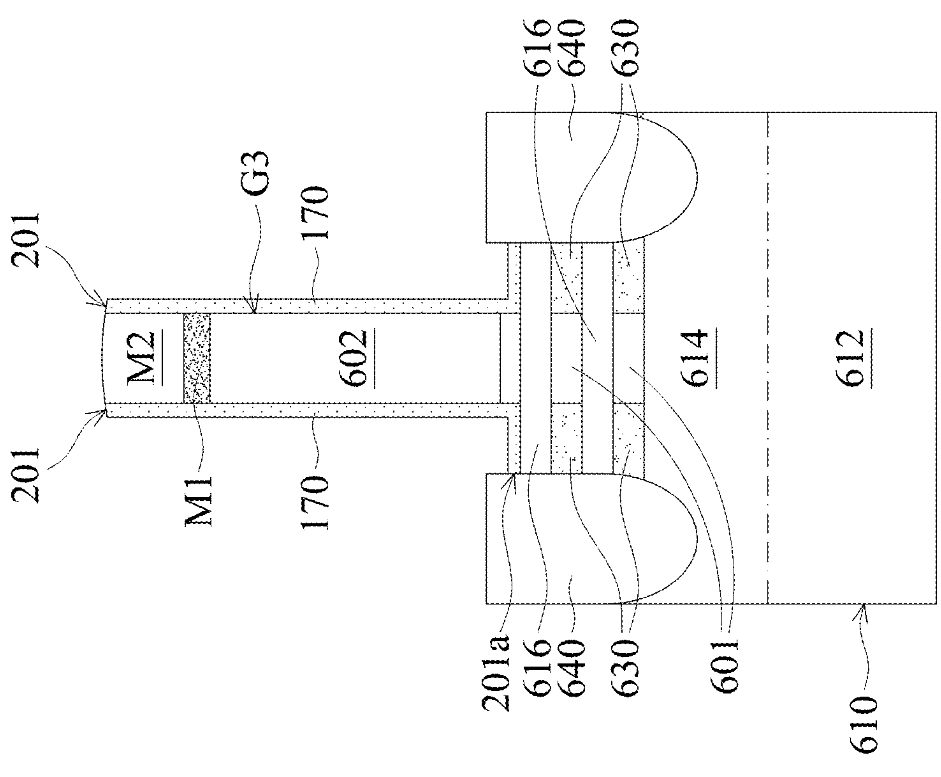

FIG. 7C-1

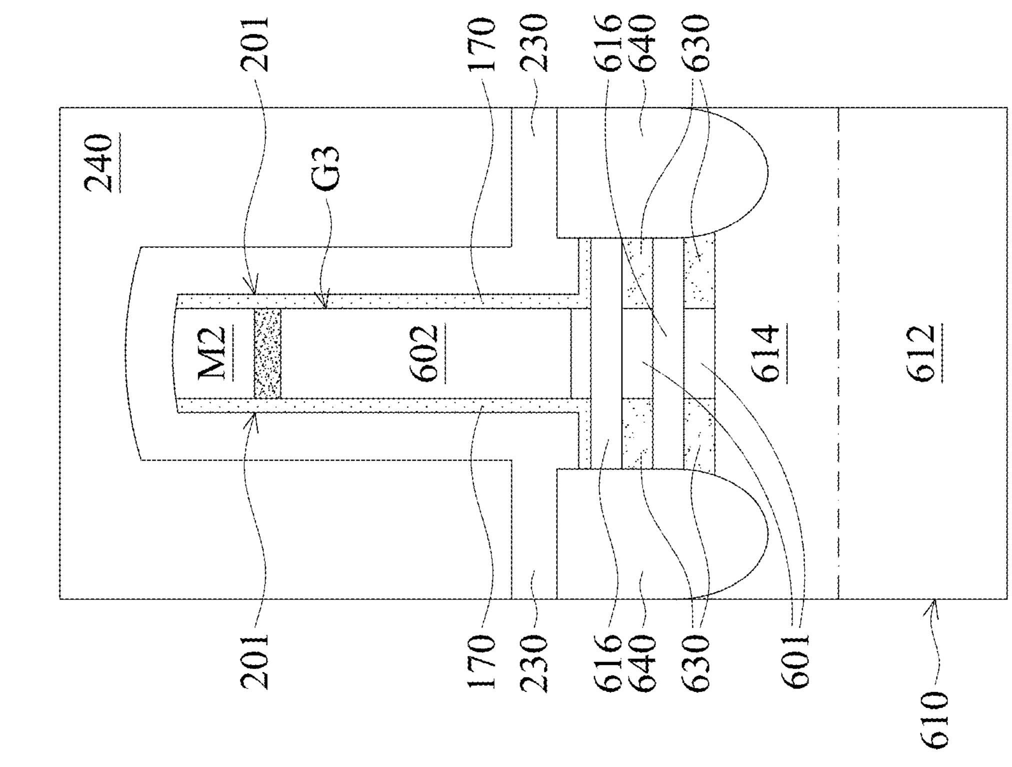

FIG. 7D-1

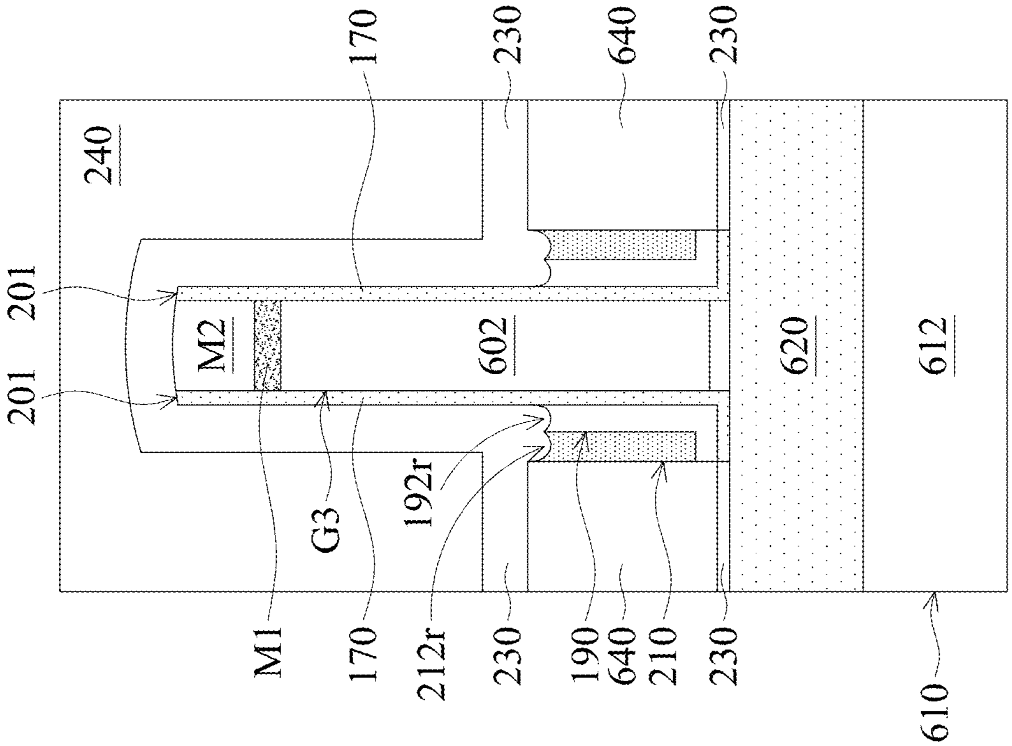

FIG. 7D-2

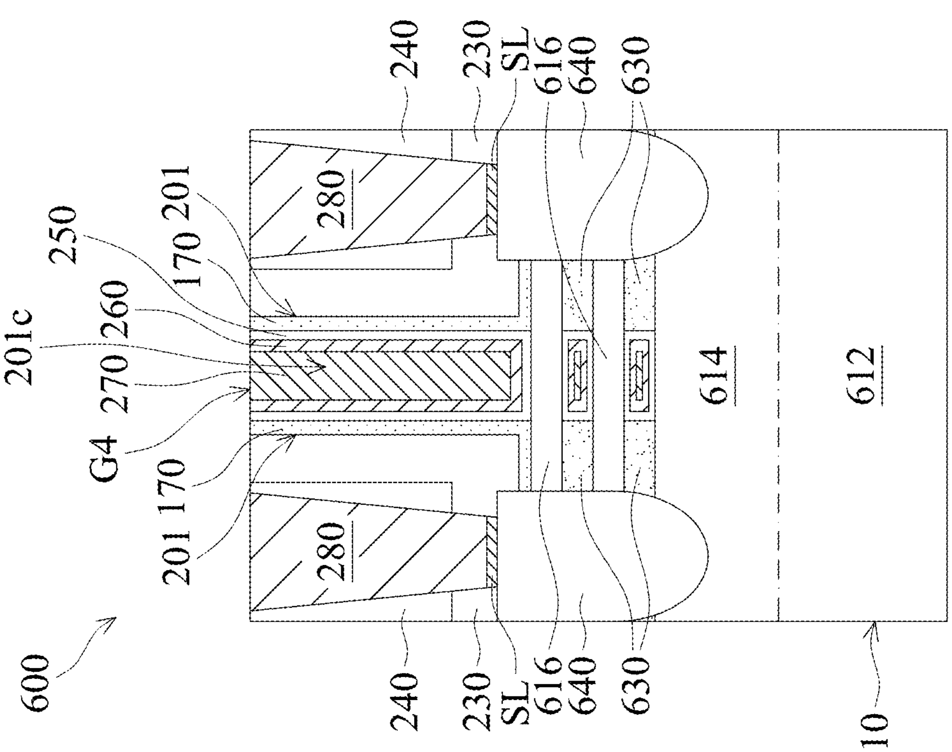

FIG. 7E-1

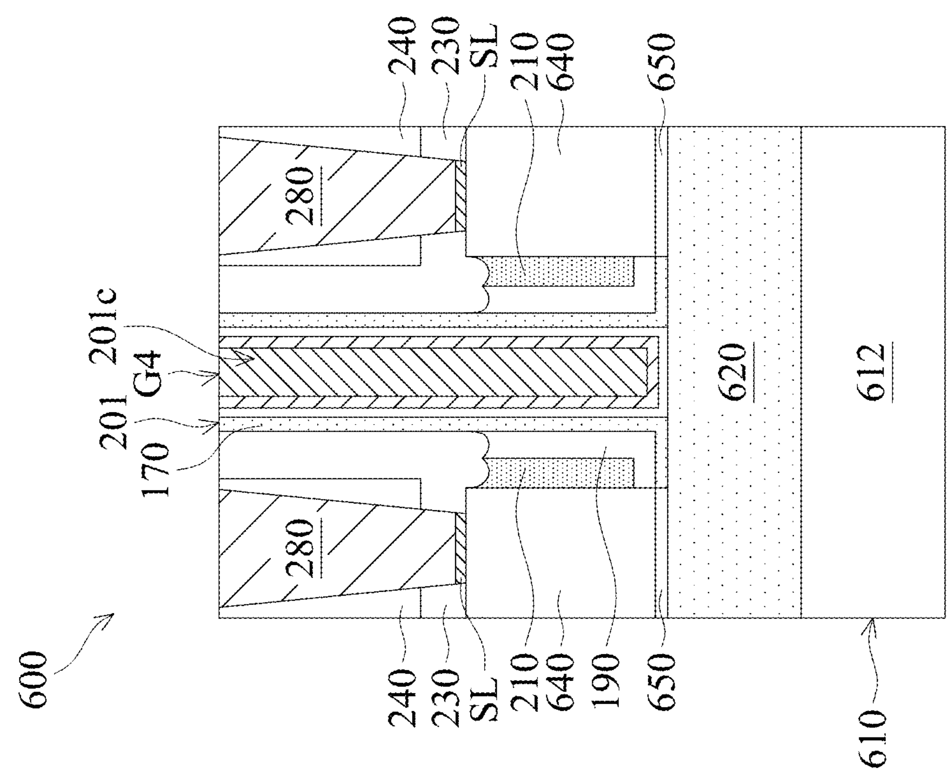

FIG. 7E-2

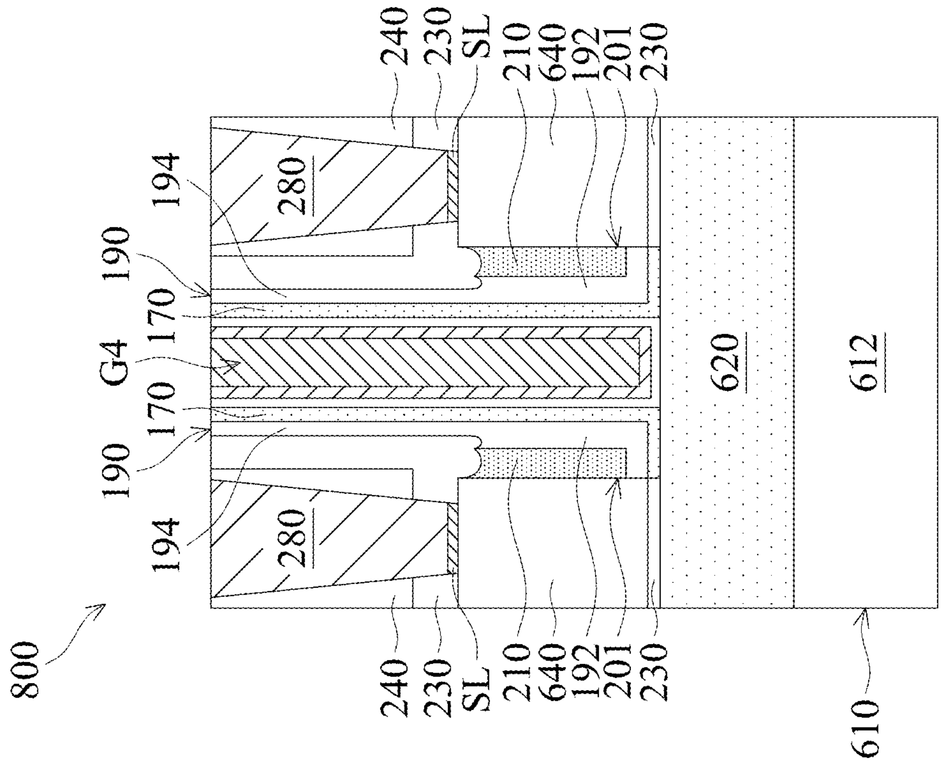
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FIG. 8B

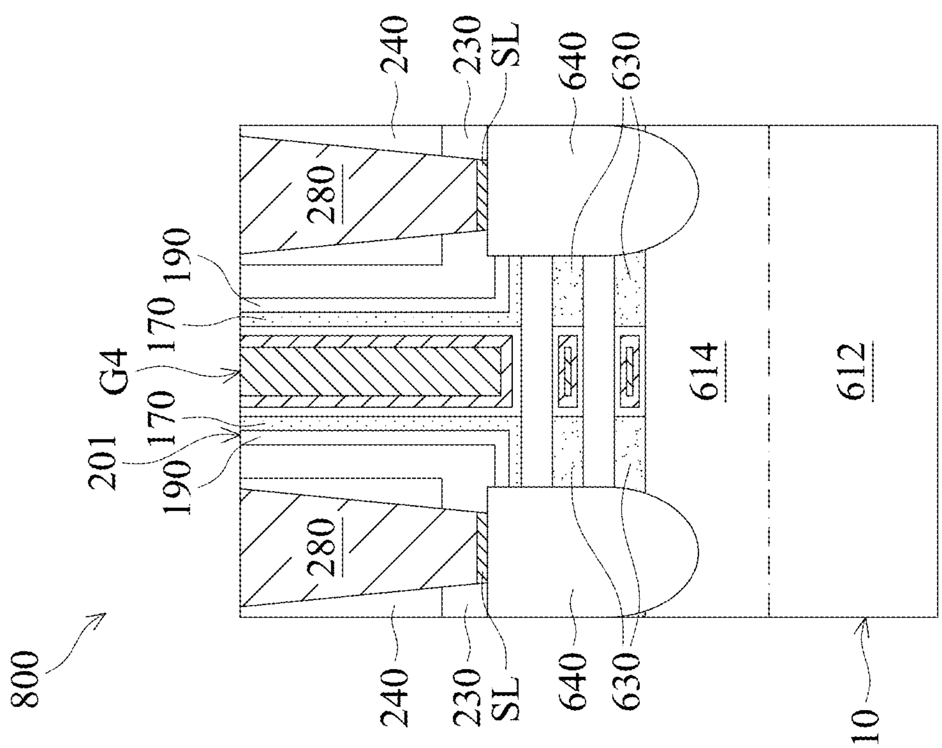
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FIG. 8A

SEMICONDUCTOR DEVICE STRUCTURE AND METHOD FOR FORMING THE SAME

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a divisional of and claims priority to U.S. patent application Ser. No. 17/717,892, titled "SEMICONDUCTOR DEVICE STRUCTURE AND METHOD FOR FORMING THE SAME" and filed on Apr. 11, 2022, which is incorporated herein by reference.

BACKGROUND

[0002] The semiconductor integrated circuit (IC) industry has experienced rapid growth. Technological advances in IC materials and design have produced generations of ICs. Each generation has smaller and more complex circuits than the previous generation. However, these advances have increased the complexity of processing and manufacturing ICs.

[0003] In the course of IC evolution, functional density (i.e., the number of interconnected devices per chip area) has generally increased while geometric size (i.e., the smallest component (or line) that can be created using a fabrication process) has decreased. This scaling-down process generally provides benefits by increasing production efficiency and lowering associated costs.

[0004] However, since feature sizes continue to decrease, fabrication processes continue to become more difficult to perform. Therefore, it is a challenge to form reliable semiconductor devices at smaller and smaller sizes. It is also a challenge to improve the performance of semiconductor devices at smaller and smaller sizes.

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It should be noted that, in accordance with standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0006] FIG. **1** is a perspective view of a semiconductor device structure, in accordance with some embodiments.

[0007] FIGS. **2**A-**2**C are cross-sectional views of various stages of a process for forming a semiconductor device structure, in accordance with some embodiments.

[0008] FIG. **3** is a perspective view of a semiconductor device structure, in accordance with some embodiments.

[0009] FIGS. **4**A-**1** to **4**H-**1** are cross-sectional views of various stages of a process for forming a semiconductor device structure, in accordance with some embodiments.

[0010] FIGS. **4**A-**2** to **4**H-**2** are cross-sectional views of various stages of a process for forming a semiconductor device structure, in accordance with some embodiments.

[0011] FIG. **5**A is a cross-sectional view of a semiconductor device structure, in accordance with some embodiments.

[0012] FIG. **5**B is a cross-sectional view of a semiconductor device structure, in accordance with some embodiments.

[0013] FIG. **6** is a perspective view of a semiconductor device structure, in accordance with some embodiments.

[0014] FIGS. **7**A-**1** to **7**E-**1** are cross-sectional views of various stages of a process for forming a semiconductor device structure, in accordance with some embodiments.

[0015] FIGS. **7**A-**2** to **7**E-**2** are cross-sectional views of various stages of a process for forming a semiconductor device structure, in accordance with some embodiments.

[0016] FIG. **8**A is a cross-sectional view of a semiconductor device structure, in accordance with some embodiments.

[0017] FIG. **8**B is a cross-sectional view of a semiconductor device structure, in accordance with some embodiments.

DETAILED DESCRIPTION

[0018] The following disclosure provides many different embodiments, or examples, for implementing different features of the subject matter provided. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0019] Furthermore, spatially relative terms, such as "beneath," "below," "lower," "above," "upper" and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0020] The term "substantially" in the description, such as in "substantially flat" or in "substantially coplanar", etc., will be understood by the person skilled in the art. In some embodiments the adjective substantially may be removed. Where applicable, the term "substantially" may also include embodiments with "entirely", "completely", "all", etc. The term "substantially" may be varied in different technologies and be in the deviation range understood by the skilled in the art. For example, the term "substantially" may also relate to 90% of what is specified or higher, such as 95% of what is specified or higher, especially 99% of what is specified or higher, including 100% of what is specified, though the present invention is not limited thereto. Furthermore, terms such as "substantially parallel" or "substantially perpendicular" may be interpreted as not to exclude insignificant deviation from the specified arrangement and may include for example deviations of up to 10°. The word "substantially" does not exclude "completely" e.g. a composition which is "substantially free" from Y may be completely free from Y.

[0021] The term "about" may be varied in different technologies and be in the deviation range understood by the skilled in the art. The term "about" in conjunction with a specific distance or size is to be interpreted so as not to exclude insignificant deviation from the specified distance or size. For example, the term "about" may include deviations of up to 10% of what is specified, though the present invention is not limited thereto. The term "about" in relation to a numerical value x may mean x±5 or 10% of what is specified, though the present invention is not limited thereto.

[0022] Some embodiments of the disclosure are described. Additional operations can be provided before, during, and/or after the stages described in these embodiments. Some of the stages that are described can be replaced or eliminated for different embodiments. Additional features can be added to the semiconductor device structure. Some of the features described below can be replaced or eliminated for different embodiments. Although some embodiments are discussed with operations performed in a particular order, these operations may be performed in another logical order.

[0023] Embodiments of the disclosure form a semiconductor device structure with FinFETs. The fins may be patterned by any suitable method. For example, the fins may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in one embodiment, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers may then be used to pattern the fins.

[0024] The gate all around (GAA) transistor structures may be patterned by any suitable method. For example, the structures may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using single, direct photolithography process. For example, in one embodiment, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers may then be used to pattern the GAA structure.

[0025] FIG. **1** is a perspective view of a semiconductor device structure, in accordance with some embodiments. FIGS. **2A-2C** are cross-sectional views of various stages of a process for forming a semiconductor device structure, in accordance with some embodiments. FIG. **2A** is a cross-sectional view illustrating the semiconductor device structure along a sectional line I-I' in FIG. **1**, in accordance with some embodiments.

[0026] As shown in FIGS. **1** and **2A**, a substrate **110** is provided, in accordance with some embodiments. The substrate **110** has a base portion **112** and a fin portion **114**, in accordance with some embodiments. The fin portion **114** is over the base portion **112**, in accordance with some embodiments.

[0027] The substrate **110** includes, for example, a semiconductor substrate. The substrate **110** includes, for example, a semiconductor wafer (such as a silicon wafer) or a portion of a semiconductor wafer. In some embodiments, the substrate **110** is made of an elementary semiconductor material including silicon or germanium in a single crystal structure, a polycrystal structure, or an amorphous structure.

[0028] In some other embodiments, the substrate **110** is made of a compound semiconductor, such as silicon carbide, gallium arsenide, gallium phosphide, indium phosphide, indium arsenide, an alloy semiconductor, such as SiGe or GaAsP, or a combination thereof. The substrate **110** may also include multi-layer semiconductors, semiconductor on insulator (SOI) (such as silicon on insulator or germanium on insulator), or a combination thereof.

[0029] In some embodiments, the substrate **110** is a device wafer that includes various device elements. In some embodiments, the various device elements are formed in and/or over the substrate **110**. The device elements are not shown in figures for the purpose of simplicity and clarity. Examples of the various device elements include active devices, passive devices, other suitable elements, or a combination thereof. The active devices may include transistors or diodes (not shown) formed at a surface of the substrate **110**. The passive devices include resistors, capacitors, or other suitable passive devices.

[0030] For example, the transistors may be metal oxide semiconductor field effect transistors (MOSFET), complementary metal oxide semiconductor (CMOS) transistors, bipolar junction transistors (BJT), high-voltage transistors, high-frequency transistors, p-channel and/or n-channel field effect transistors (PFETs/NFETs), etc. Various processes, such as front-end-of-line (FEOL) semiconductor fabrication processes, are performed to form the various device elements. The FEOL semiconductor fabrication processes may include deposition, etching, implantation, photolithography, annealing, planarization, one or more other applicable processes, or a combination thereof.

[0031] In some embodiments, isolation features (not shown) are formed in the substrate **110**. The isolation features are used to surround active regions and electrically isolate various device elements formed in and/or over the substrate **110** in the active regions. In some embodiments, the isolation features include shallow trench isolation (STI) features, local oxidation of silicon (LOCOS) features, other suitable isolation features, or a combination thereof.

[0032] As shown in FIGS. **1** and **2A**, an isolation layer **120** is formed over the base portion **112**, in accordance with some embodiments. The fin portion **114** is partially in the isolation layer **120**, in accordance with some embodiments. The isolation layer **120** surrounds a lower portion of the fin portion **114**, in accordance with some embodiments. The isolation layer **120** includes oxide (such as silicon oxide), in accordance with some embodiments. The isolation layer **120** is formed by a chemical vapor deposition (CVD) process and an etching back process, in accordance with some embodiments.

[0033] As shown in FIGS. **1** and **2A**, a gate stack G**1** and mask layers **150** and **160** are formed over the fin portion **114** and the isolation layer **120**, in accordance with some embodiments. The gate stack G**1** is wrapped around an upper part of the fin portion **114**, in accordance with some embodiments.

[0034] The gate stack G**1** includes a gate dielectric layer **130** and a gate electrode **140**, in accordance with some embodiments. The gate dielectric layer **130** is formed over the fin portion **114** and the isolation layer **120**, in accordance with some embodiments. The gate electrode **140** is formed over the gate dielectric layer **130**, in accordance with some embodiments.

[0035] The gate dielectric layer **130** is made of an insulating material, such as oxide (e.g., silicon oxide), in accordance with some embodiments. The gate electrode **140** is made of a semiconductor material (e.g. polysilicon) or a conductive material (e.g., metal or alloy), in accordance with some embodiments.

[0036] The formation of the gate dielectric layer **130** and the gate electrode **140** includes: depositing a gate dielectric material layer (not shown) over the fin portion **114** and the isolation layer **120**; depositing a gate electrode material layer (not shown) over the gate dielectric material layer; sequentially forming the mask layers **150** and **160** over the gate electrode material layer, wherein the mask layers **150** and **160** expose portions of the gate electrode material layer; and removing the exposed portions of the gate electrode material layer and the gate dielectric material layer thereunder, in accordance with some embodiments.

[0037] In some embodiments, the mask layer **150** serves as a buffer layer or an adhesion layer that is formed between the underlying gate electrode **140** and the overlying mask layer **160**. The mask layer **150** may also be used as an etch stop layer when the mask layer **160** is removed or etched, in accordance with some embodiments. The mask layers **150** and **160** are made of different materials, in accordance with some embodiments.

[0038] In some embodiments, the mask layer **150** is made of oxide, such as silicon oxide. In some embodiments, the mask layer **150** is formed by a deposition process, such as a chemical vapor deposition (CVD) process, a low-pressure chemical vapor deposition (LPCVD) process, a plasma enhanced chemical vapor deposition (PECVD) process, a high-density plasma chemical vapor deposition (HDPCVD) process, a spin-on process, or another applicable process.

[0039] In some embodiments, the mask layer **160** is made of nitride (e.g., silicon nitride), oxynitride (e.g., silicon oxynitride), or another applicable material. In some embodiments, the mask layer **160** is formed by a deposition process, such as a chemical vapor deposition (CVD) process, a low-pressure chemical vapor deposition (LPCVD) process, a plasma enhanced chemical vapor deposition (PECVD) process, a high-density plasma chemical vapor deposition (HDPCVD) process, a spin-on process, or another applicable process.

[0040] After the deposition processes for forming the mask layers **150** and **160**, the mask layers **150** and **160** are patterned by a photolithography process and an etching process, so as to expose the portions of the gate electrode material layer.

[0041] As shown in FIG. **2**B, an inner material layer **170***a* is deposited over the mask layers **150** and **160**, the gate stack G**1**, and the substrate **110**, in accordance with some embodiments. The inner material layer **170***a* conformally covers sidewalls S**1** of the gate stack G**1**, in accordance with some embodiments. The inner material layer **170***a* is configured to define the positions of doped regions and source/drain structures formed in subsequent processes, in accordance with some embodiments.

[0042] The inner material layer **170***a* is made of a dielectric material such as oxides (e.g., Al_2O_3, SiCO or SiCON) or nitrides (SiCN, SiON or SiN), in accordance with some embodiments. The inner material layer **170***a* is formed using a deposition process, such as a chemical vapor deposition (CVD) process, an atomic layer deposition (ALD) process, or a physical vapor deposition (PVD) process, in accordance with some embodiments.

[0043] Thereafter, as shown in FIG. **2**B, a doping process **180** is performed over the substrate **110** to form doped regions **114***a* in the substrate **110**, in accordance with some embodiments. The doped regions **114***a* are also referred to as lightly doped drain (LDD) regions, in accordance with some embodiments. The doped regions **114***a* are partially under the gate stack G**1**, in accordance with some embodiments.

[0044] The doping process **180** includes an ion implantation process, in accordance with some embodiments. The dopants used by the doping process **180** include N-type dopants, such as phosphorus (P) or arsenic (As), or P-type dopants, such as boron (B), in accordance with some embodiments.

[0045] Thereafter, as shown in FIG. **2**C, a middle material layer **190***a* is deposited over the inner material layer **170***a*, in accordance with some embodiments. The middle material layer **190***a* conformally covers the inner material layer **170***a*, in accordance with some embodiments. The middle material layer **190***a* is configured to define the position of source/drain structures formed in a subsequent process, in accordance with some embodiments.

[0046] The middle material layer **190***a* is made of a dielectric material such as oxides (e.g., Al_2O_3, SiCO or SiCON) or nitrides (SiCN, SiON or SiN), in accordance with some embodiments. In some embodiments, the inner material layer **170***a* and the middle material layer **190***a* are made of different materials. The middle material layer **190***a* is formed using a deposition process, such as a chemical vapor deposition (CVD) process, an atomic layer deposition (ALD) process, or a physical vapor deposition (PVD) process, in accordance with some embodiments.

[0047] As shown in FIG. **2**C, an outer material layer **210***a* is deposited over the middle material layer **190***a*, in accordance with some embodiments. The outer material layer **210***a* conformally covers the middle material layer **190***a*, in accordance with some embodiments. The outer material layer **210***a* is configured to define the position of source/drain structures formed in a subsequent process, in accordance with some embodiments.

[0048] The outer material layer **210***a* is made of a dielectric material such as oxides (e.g., Al_2O_3, SiCO or SiCON) or nitrides (SiCN, SiON or SiN), in accordance with some embodiments. The outer material layer **210***a* and the middle material layer **190***a* are made of different materials, in accordance with some embodiments.

[0049] The dielectric constant of the outer material layer **210***a* is greater than the dielectric constant of the middle material layer **190***a*, in accordance with some embodiments. The dielectric constant of the middle material layer **190***a* is greater than or equal to the dielectric constant of the inner material layer **170***a*, in accordance with some embodiments.

[0050] The outer material layer **210***a* is formed using a deposition process, such as a chemical vapor deposition (CVD) process, an atomic layer deposition (ALD) process, or a physical vapor deposition (PVD) process, in accordance with some embodiments.

[0051] FIG. **3** is a perspective view of a semiconductor device structure, in accordance with some embodiments. FIG. **4A-1** is a cross-sectional view illustrating the semiconductor device structure along a sectional line I-I' in FIG. **3**, in accordance with some embodiments. FIG. **4A-2** is a cross-sectional view illustrating the semiconductor device structure along a sectional line II-II' in FIG. **3**, in accordance with some embodiments.

[0052] As shown in FIGS. **2C**, **3**, **4A-1**, and **4A-2**, the outer material layer **210***a*, the middle material layer **190***a*, and the inner material layer **170***a* over a top surface **162** of the mask layer **160** and a top surface **121** of the isolation layer **120** are partially removed, in accordance with some embodiments.

[0053] The remaining outer material layer **210***a* forms an outer layer **210**, in accordance with some embodiments. The remaining middle material layer **190***a* forms a middle layer **190**, in accordance with some embodiments. The remaining inner material layer **170***a* forms an inner layer **170**, in accordance with some embodiments. The inner layer **170**, the middle layer **190**, and the outer layer **210** together form a spacer structure **201**, in accordance with some embodiments. The spacer structure **201** surrounds the gate stack G**1**, in accordance with some embodiments.

[0054] As shown in FIGS. **3** and **4A-1**, portions of the fin portion **114** adjacent to the spacer structure **201** are removed, in accordance with some embodiments. After the removal process, as shown in FIGS. **3** and **4A-1**, recesses **114***a* are formed in the fin portion **114**, in accordance with some embodiments. The recesses **114***a* are configured to accommodate source/drain structures formed in a subsequent process, in accordance with some embodiments. The removal process includes an etching process, such as an anisotropic etching process (e.g., a dry etching process), in accordance with some embodiments.

[0055] As shown in FIGS. **3**, **4A-1**, and **4A-2**, source/drain structures **220** are formed in the recesses **114***a* and over the fin portion **114**, in accordance with some embodiments. In some embodiments, a lower portion **201***a* of the spacer structure **201** is between the source/drain structures **220** and the gate stack G**1**.

[0056] The source/drain structures **220** are in direct contact with the fin portion **114**, in accordance with some embodiments. The source/drain structures **220** are positioned on two opposite sides of the gate stack G**1**, in accordance with some embodiments. The source/drain structures **220** include a source structure and a drain structure, in accordance with some embodiments.

[0057] The source/drain structures **220** are made of an N-type conductivity material, in accordance with some embodiments. The N-type conductivity material includes silicon phosphorus (SiP) or another suitable N-type conductivity material.

[0058] The source/drain structures **220** are doped with the Group VA element, in accordance with some embodiments. The Group VA element includes phosphor (P), antimony (Sb), or another suitable Group VA material. In some embodiments, a concentration of the Group VA element (e.g. phosphor) ranges from about 3E21 atoms/cm^3 to about 7E21 atoms/cm^3. The source/drain structures **220** are also referred to as doped structures, in accordance with some embodiments.

[0059] In some other embodiments, the source/drain structures **220** are made of a P-type conductivity material, in accordance with some embodiments. The P-type conductivity material includes silicon germanium (SiGe) or another suitable P-type conductivity material.

[0060] The source/drain structures **220** are doped with the Group IIIA element, in accordance with some embodiments. The Group IIIA element includes boron or another suitable material. The source/drain structures **220** are formed using an epitaxial process, in accordance with some embodiments.

[0061] FIGS. **4A-1** to **4H-1** are cross-sectional views of various stages of a process for forming a semiconductor device structure, in accordance with some embodiments. FIGS. **4A-2** to **4H-2** are cross-sectional views of various stages of a process for forming a semiconductor device structure, in accordance with some embodiments.

[0062] Thereafter, as shown in FIGS. **4B-1** and **4B-2**, an upper portion of the outer layer **210** above the source/drain structures **220** is removed, in accordance with some embodiments. In some embodiments, a lower portion **212** of the outer layer **210** remains between the source/drain structures **220** and the gate stack G**1**. The lower portion **212** has a top surface **212***a*, in accordance with some embodiments.

[0063] As shown in FIGS. **4B-1** and **4B-2**, a recess **212***r* is formed in the top surface **212***a* of the lower portion **212**, in accordance with some embodiments. The top surface **212***a* is lower than top surfaces **222** of the source/drain structures **220**, in accordance with some embodiments. The removal process includes an isotropic etching process, such as a wet etching process, in accordance with some embodiments. In some embodiments, the outer layer **210** is made of silicon nitride, and the etchant used in the wet etching process includes phosphoric acid.

[0064] Afterwards, as shown in FIGS. **4C-1** and **4C-2**, an upper portion of the middle layer **190** above the source/drain structures **220** is removed, in accordance with some embodiments. As shown in FIG. **4C-2**, a lower portion **192** of the middle layer **190** remains between the outer layer **210** and the gate stack G**1**, in accordance with some embodiments.

[0065] The lower portion **192** has a top surface **192***a*, in accordance with some embodiments. The top surface **192***a* is lower than the top surface **222** of the source/drain structure **220**, in accordance with some embodiments. As shown in FIG. **4C-2**, a recess **192***r* is formed in the top surface **192***a* of the lower portion **192**, in accordance with some embodiments.

[0066] The recess **212***r* in the top surface **212***a* of the outer layer **210** and the recess **192***r* in the top surface **192***a* of the middle layer **190** together form a recess R**1**, in accordance with some embodiments. The recess R**1** has a W-like shape, in accordance with some embodiments. The depth D**1** of the recess **212***r* ranges from about 1 nm to about 6 nm, in accordance with some embodiments. The depth D**2** of the recess **192***r* ranges from about 1 nm to about 6 nm, in accordance with some embodiments.

[0067] The recess **212***r* has opposite edges **212**E**1** and **212**E**2**, in accordance with some embodiments. The edge **212**E**1** is adjacent to the source/drain structure **220**, in accordance with some embodiments. The edge **212**E**2** is adjacent to the middle layer **190**, in accordance with some embodiments. The edge **212**E**1** is higher than the edge **212**E**2**, in accordance with some embodiments.

[0068] The recess **192***r* has edges **192**E**1** and **192**E**2**, in accordance with some embodiments. The edge **192**E**1** is adjacent to the gate stack G**1**, in accordance with some embodiments. The edge **192**E**2** is adjacent to the outer layer

210, in accordance with some embodiments. The edge **192**E**1** is higher than the edge **192**E**2**, in accordance with some embodiments.

[0069] The lower portion **201**a of the spacer structure **201** is thicker than an upper portion **201**b of the spacer structure **201**, in accordance with some embodiments. The thickness T**201**a of the lower portion **201**a is greater than the thickness T**201**b of the upper portion **201**b, in accordance with some embodiments. The spacer structure **201** has an L-like shape, in accordance with some embodiments.

[0070] Thereafter, as shown in FIGS. **4**D-**1** and **4**D-**2**, an etch stop layer **230** is formed over the source/drain structure **220**, the spacer structure **201**, and the mask layer **160**, in accordance with some embodiments. The etch stop layer **230** is filled into the recess **192**r of the middle layer **190** and the recess **212**r of the outer layer **210**, in accordance with some embodiments.

[0071] The etch stop layer **230** is made of an insulating material, such as a nitrogen-containing material (e.g., silicon nitride), in accordance with some embodiments. The etch stop layer **230** is formed using a deposition process, such as a chemical vapor deposition (CVD) process, an atomic layer deposition (ALD) process, or a physical vapor deposition (PVD) process, in accordance with some embodiments.

[0072] Afterwards, as shown in FIGS. **4**D-**1** and **4**D-**2**, a dielectric layer **240** is formed over the etch stop layer **230**, in accordance with some embodiments. The dielectric layer **240** is made of any suitable insulating material, such as silicon oxide, silicon oxynitride, borosilicate glass (BSG), phosphoric silicate glass (PSG), borophosphosilicate glass (BPSG), fluorinated silicate glass (FSG), low-k material, porous dielectric material, or a combination thereof. The dielectric layer **240** is deposited by any suitable process, such as a CVD process, a spin-on process, a sputtering process, or a combination thereof, in accordance with some embodiments.

[0073] Afterwards, as shown in FIGS. **4**D-**1**, **4**E-**1** and **4**E-**2**, a planarization process is performed on the dielectric layer **240** until a top surface **142** of the gate electrode **140** is exposed, in accordance with some embodiments. After the planarization process is performed, the mask layers **150** and **160** and upper portions of the dielectric layer **240**, the etch stop layer **230**, the spacer structure **201**, and the gate stack G**1** are removed, in accordance with some embodiments.

[0074] After the planarization process, top surfaces **242**, **232**, **172**, and **142** of the dielectric layer **240**, the etch stop layer **230**, the inner layer **170**, and the gate electrode **140** are substantially level with each other or substantially coplanar with each other, in accordance with some embodiments. The planarization process includes a chemical mechanical polishing (CMP) process, in accordance with some embodiments.

[0075] Thereafter, a gate replacement process is performed, in accordance with some embodiments. As shown in FIGS. **4**F-**1** and **4**F-**2**, the gate stack G**1** is removed, in accordance with some embodiments. The removal process includes a wet etching process, in accordance with some embodiments. After the removal process, a trench **201**c is formed in the spacer structure **201**, in accordance with some embodiments. The trench **201**c exposes the fin portion **114**, in accordance with some embodiments.

[0076] As shown in FIGS. **4**F-**1** and **4**F-**2**, a gate dielectric material layer **250**a is conformally formed in the trench **201**c, in accordance with some embodiments. The gate dielectric material layer **250**a is made of a dielectric material, such as a high dielectric constant (high-k) material, in accordance with some embodiments.

[0077] The high-k material is made of metal oxides, such as hafnium oxide (HfO_2), hafnium silicon oxide (HfSiO), hafnium silicon oxynitride (HfSiON), hafnium tantalum oxide (HfTaO), hafnium titanium oxide (HfTiO), hafnium zirconium oxide (HfZrO), aluminum oxide, hafnium dioxide-alumina (HfO_2—Al_2O_3) alloy, other suitable high-k dielectric materials, or combinations thereof, in accordance with some embodiments. In some other embodiments, the high-k material is made of metal nitrides, metal silicates, transition metal-oxides, transition metal-nitrides, transition metal-silicates, oxynitrides of metals, other suitable materials, or combinations thereof.

[0078] As shown in FIGS. **4**F-**1** and **4**F-**2**, a work function metal material layer **260**a is conformally formed over the gate dielectric material layer **250**a, in accordance with some embodiments. The work function metal material layer **260**a conformally covers the gate dielectric material layer **250**a, in accordance with some embodiments. The work function metal material layer **260**a provides a desired work function for transistors to enhance device performance including improved threshold voltage.

[0079] In the embodiments of forming an NMOS transistor, the work function metal material layer **260**a can be an n-type metal capable of providing a work function value suitable for the device, such as equal to or less than about 4.5 eV. The n-type metal may be made of metal, metal carbide, metal nitride, or a combination thereof. For example, the n-type metal is made of tantalum, tantalum nitride, or a combination thereof.

[0080] In the embodiments of forming a PMOS transistor, the work function metal material layer **260**a can be a p-type metal capable of providing a work function value suitable for the device, such as equal to or greater than about 4.8 eV. The p-type metal may be made of metal, metal carbide, metal nitride, other suitable materials, or a combination thereof. For example, the p-type metal is made of titanium, titanium nitride, other suitable materials, or a combination thereof.

[0081] The work function metal material layer **260**a is formed using a deposition process, in accordance with some embodiments. The deposition process includes a physical vapor deposition (PVD) process, a CVD process, an atomic layer deposition (ALD), a plating process, another suitable method, or a combination thereof.

[0082] Afterwards, as shown in FIGS. **4**F-**1** and **4**F-**2**, a gate electrode material layer **270**a is deposited over the work function metal material layer **260**a to fill the trench **201**c of the spacer structure **201**, in accordance with some embodiments. The gate electrode material layer **270**a is made of a suitable metal material, such as aluminum, tungsten, gold, platinum, cobalt, another suitable metal, an alloy thereof, or a combination thereof, in accordance with some embodiments.

[0083] Afterwards, as shown in FIGS. **4**G-**1** and **4**G-**2**, portions of the gate electrode material layer **270**a, the work function metal material layer **260**a, and the gate dielectric material layer **250**a outside of the trench **201**c of the spacer structure **201** are removed, in accordance with some embodiments.

[0084] The remaining gate electrode material layer **270**a forms a gate electrode layer **270**, in accordance with some embodiments. The remaining work function metal material layer **260***a* forms a work function metal layer **260**, in accordance with some embodiments. The remaining gate dielectric material layer **250***a* forms a gate dielectric layer **250**, in accordance with some embodiments.

[0085] The gate electrode layer **270**, the work function metal layer **260**, and the gate dielectric layer **250** together form a gate stack G**2**, in accordance with some embodiments. The gate stack G**2** and the source/drain structures **220** together form a transistor TR**1**, in accordance with some embodiments.

[0086] Thereafter, as shown in FIGS. **4**H-**1** and **4**H-**2**, portions of the dielectric layer **240** and the etch stop layer **230** are removed to form contact holes H in the dielectric layer **240** and the etch stop layer **230**, in accordance with some embodiments. The contact holes H pass though the dielectric layer **240** and the etch stop layer **230**, in accordance with some embodiments. The contact hole H partially exposes the top surface **222** of the source/drain structure **220** thereunder, in accordance with some embodiments.

[0087] Afterwards, a metal layer (not shown) is formed over the source/drain structures **220**, in accordance with some embodiments. The metal layer is made of Ti, Co, Ru, or another suitable metal material. The metal layer is formed using a deposition process, such as a physical vapor deposition process, a plating process, another suitable method, or a combination thereof, in accordance with some embodiments.

[0088] Thereafter, as shown in FIGS. **4**H-**1** and **4**H-**2**, the metal layer and the source/drain structures **220** are annealed to react the metal layer with the source/drain structures **220** so as to form a silicide layer SL between the metal layer and the source/drain structures **220**, in accordance with some embodiments. The silicide layer SL is in direct contact with the source/drain structures **220**, in accordance with some embodiments. The silicide layer SL includes $TiSi_2$ (titanium disilicide), $CoSi_2$, or RuSi, in accordance with some embodiments.

[0089] As shown in FIGS. **4**H-**1** and **4**H-**2**, the metal layer, which has not reacted with the source/drain structures **220**, is removed, in accordance with some embodiments. The removal process includes an etching process such as a wet etching process or a dry etching process, in accordance with some embodiments.

[0090] As shown in FIGS. **4**H-**1** and **4**H-**2**, contact structures **280** are formed in the contact holes H, in accordance with some embodiments. The contact structures **280** pass through the dielectric layer **240** and the etch stop layer **230** to connect to the silicide layer SL, in accordance with some embodiments.

[0091] The formation of the contact structures **280** includes depositing a conductive material layer (not shown) over the dielectric layer **240** and in the contact holes H; and performing a chemical mechanical polishing (CMP) process over the conductive material layer to remove the conductive material layer outside of the contact holes H, in accordance with some embodiments.

[0092] The contact structures **280** are made of tungsten (W) or another suitable conductive material, in accordance with some embodiments. In this step, a semiconductor device structure **400** is substantially formed, in accordance with some embodiments.

[0093] Since the embodiment removes the upper portions of the middle layer **190** and the outer layer **210** to enlarge the space for accommodating the contact structures **280**, in accordance with some embodiments. Therefore, the yield of the formation of the contact structures **280** is improved, in accordance with some embodiments. Furthermore, the width of the contact structure **280** is able to be enlarged, which reduces the contact resistance between the contact structure **280** and the source/drain structures **220**, in accordance with some embodiments. Therefore, the performance of the semiconductor device structure **400** is improved, in accordance with some embodiments.

[0094] Since the lower portions of the middle layer **190** and the outer layer **210** remain between the gate stack G**2** and the source/drain structures **220**, the distance between the gate stack G**2** and the source/drain structures **220** is maintained, in accordance with some embodiments. Therefore, the capacitance between the gate stack G**2** and the source/drain structures **220** is maintained, in accordance with some embodiments.

[0095] Furthermore, the width of the contact structure **280** is able to be maintained when the distance between the gate stack G**2** and the adjacent gate stack (not shown) is reduced, which helps the semiconductor device structure **400** to be scaled down, in accordance with some embodiments.

[0096] In addition, the embodiment removes the upper portions of the middle layer **190** and the outer layer **210** without using masks, which does not increase the cost of the process for forming the semiconductor device structure **400** and increases process compatibility, in accordance with some embodiments.

[0097] FIG. **5**A is a cross-sectional view of a semiconductor device structure, in accordance with some embodiments. FIG. **5**B is a cross-sectional view of a semiconductor device structure, in accordance with some embodiments.

[0098] After the step of FIGS. **4**B-**1** and **4**B-**2**, as shown in FIGS. **5**A and **5**B, an upper portion **194** of the middle layer **190** is thinned, in accordance with some embodiments. After the thinning process, the upper portion **194** is thinner than the lower portion **192** of the middle layer **190**, in accordance with some embodiments.

[0099] The upper portion **194** is over the source/drain structures **220**, in accordance with some embodiments. The lower portion **192** is between the source/drain structures **220** and the gate stack G**1** (or the gate stack G**2**), in accordance with some embodiments.

[0100] The thinning process forms a recess **192***r* in the lower portion **192**, in accordance with some embodiments. The recess **192***r* is narrower than the lower portion **192**, in accordance with some embodiments. The recess **192***r* is narrower than the recess **212***r* in the lower portion **212** of the outer layer **210**, in accordance with some embodiments. The thinning process includes a dry etching process or another suitable etching process, in accordance with some embodiments.

[0101] Thereafter, the steps of FIGS. **4**D-**1** to **4**H-**1** and **4**D-**2** to **4**H-**2** are performed to form the etch stop layer **230**, the dielectric layer **240**, the gate stack G**2**, which includes the gate dielectric layer **250**, the work function metal layer **260**, and the gate electrode layer **270**, the silicide layer SL, and the contact structures **280**, in accordance with some embodiments. In this step, a semiconductor device structure **500** is substantially formed, in accordance with some embodiments.

[0102] FIG. **6** is a perspective view of a semiconductor device structure **600**, in accordance with some embodiments. FIG. **7A-1** is a cross-sectional view illustrating the semiconductor device structure along a sectional line I-I' in FIG. **6**, in accordance with some embodiments. FIG. **7A-2** is a cross-sectional view illustrating the semiconductor device structure **600** along a sectional line II-II' in FIG. **6**, in accordance with some embodiments.

[0103] FIGS. **7A-1** to **7E-1** are cross-sectional views of various stages of a process for forming a semiconductor device structure, in accordance with some embodiments. FIGS. **7A-2** to **7E-2** are cross-sectional views of various stages of a process for forming a semiconductor device structure, in accordance with some embodiments.

[0104] As shown in FIGS. **6**, **7A-1**, and **7A-2**, a semiconductor device structure **600** is provided, in accordance with some embodiments. The semiconductor device structure **600** includes a substrate **610**, nanostructures **601**, an isolation layer **620**, a gate stack G**3**, a spacer structure **201**, an inner spacer **630**, source/drain structures **640**, and mask layers M**1** and M**2**, in accordance with some embodiments.

[0105] The substrate **610** has a base portion **612**, a fin portion **614**, and nanostructures **616**, in accordance with some embodiments. The fin portion **614** is over the base portion **612**, in accordance with some embodiments. The nanostructures **616** are over the fin portion **614**, in accordance with some embodiments.

[0106] The substrate **610** includes, for example, a semiconductor substrate. The substrate **610** includes, for example, a semiconductor wafer (such as a silicon wafer) or a portion of a semiconductor wafer. In some embodiments, the substrate **610** is made of an elementary semiconductor material including silicon or germanium in a single crystal structure, a polycrystal structure, or an amorphous structure.

[0107] In some other embodiments, the substrate **610** is made of a compound semiconductor, such as silicon carbide, gallium arsenide, gallium phosphide, indium phosphide, indium arsenide, an alloy semiconductor, such as SiGe or GaAsP, or a combination thereof. The substrate **610** may also include multi-layer semiconductors, semiconductor on insulator (SOI) (such as silicon on insulator or germanium on insulator), or a combination thereof.

[0108] In some embodiments, the substrate **610** is a device wafer that includes various device elements. In some embodiments, the various device elements are formed in and/or over the substrate **610**. The device elements are not shown in figures for the purpose of simplicity and clarity. Examples of the various device elements include active devices, passive devices, other suitable elements, or a combination thereof. The active devices may include transistors or diodes (not shown) formed at a surface of the substrate **610**. The passive devices include resistors, capacitors, or other suitable passive devices.

[0109] For example, the transistors may be metal oxide semiconductor field effect transistors (MOSFET), complementary metal oxide semiconductor (CMOS) transistors, bipolar junction transistors (BJT), high-voltage transistors, high-frequency transistors, p-channel and/or n-channel field effect transistors (PFETs/NFETs), etc.

[0110] Various processes, such as front-end-of-line (FEOL) semiconductor fabrication processes, are performed to form the various device elements. The FEOL semiconductor fabrication processes may include deposition, etching, implantation, photolithography, annealing, planarization, one or more other applicable processes, or a combination thereof.

[0111] In some embodiments, isolation features (not shown) are formed in the substrate **610**. The isolation features are used to surround active regions and electrically isolate various device elements formed in and/or over the substrate **610** in the active regions. In some embodiments, the isolation features include shallow trench isolation (STI) features, local oxidation of silicon (LOCOS) features, other suitable isolation features, or a combination thereof.

[0112] The isolation layer **620** is formed over the base portion **612**, in accordance with some embodiments. The fin portion **614** is partially in the isolation layer **620**, in accordance with some embodiments. The isolation layer **620** surrounds the fin portion **614**, in accordance with some embodiments. The isolation layer **620** includes oxide (such as silicon oxide), in accordance with some embodiments. The isolation layer **620** is formed by a chemical vapor deposition (CVD) process and an etching back process, in accordance with some embodiments.

[0113] The gate stack G**3** is formed over the fin portion **614**, the nanostructures **616**, and the isolation layer **620**, in accordance with some embodiments. The gate stack G**3** surrounds the nanostructures **601** and **616** and an upper portion of the fin portion **614**, in accordance with some embodiments.

[0114] The gate stack G**3** includes a gate electrode **602** and a gate dielectric layer **603**, in accordance with some embodiments. The gate dielectric layer **603** is formed over the nanostructures **601** and **616**, the fin portion **614**, and the isolation layer **620**, in accordance with some embodiments. The gate electrode **602** is formed over the gate dielectric layer **603**, in accordance with some embodiments.

[0115] The gate electrode **602** is made of a semiconductor material (e.g. polysilicon) or a conductive material (e.g., metal or alloy), in accordance with some embodiments. The gate dielectric layer **603** is made of an insulating material, such as oxide (e.g., silicon oxide), in accordance with some embodiments.

[0116] The mask layers M**1** and M**2** are sequentially stacked over the gate stack G**3**, in accordance with some embodiments. In some embodiments, the mask layer M**1** serves as a buffer layer or an adhesion layer that is formed between the underlying gate electrode **602** and the overlying mask layer M**2**. The mask layer M**1** may also be used as an etch stop layer when the mask layer M**2** is removed or etched, in accordance with some embodiments. The mask layers M**1** and M**2** are made of different materials, in accordance with some embodiments.

[0117] In some embodiments, the mask layer M**1** is made of oxide, such as silicon oxide. In some embodiments, the mask layer M**1** is formed by a deposition process, such as a chemical vapor deposition (CVD) process, a low-pressure chemical vapor deposition (LPCVD) process, a plasma enhanced chemical vapor deposition (PECVD) process, a high-density plasma chemical vapor deposition (HDPCVD) process, a spin-on process, or another applicable process.

[0118] In some embodiments, the mask layer M**2** is made of nitride (e.g., silicon nitride), oxynitride (e.g., silicon oxynitride), or another applicable material. In some embodiments, the mask layer M**2** is formed by a deposition process, such as a chemical vapor deposition (CVD) process, a low-pressure chemical vapor deposition (LPCVD) process, a plasma enhanced chemical vapor deposition (PECVD) process, a high-density plasma chemical vapor deposition (HDPCVD) process, a spin-on process, or another applicable process.

[0119] After the deposition processes for forming the mask layers M**1** and M**2**, the mask layers M**1** and M**2** are patterned by a photolithography process and an etching process, in accordance with some embodiments.

[0120] The spacer structure **201** is formed over sidewalls S**2** of the gate stack G**3**, the nanostructures **616**, and the isolation layer **620**, in accordance with some embodiments. The inner layer **170** conformally covers the sidewalls S**2** of the gate stack G**3**, the nanostructures **616**, and a top surface **622** of the isolation layer **620**, in accordance with some embodiments. The middle layer **190** conformally covers of the inner layer **170**, in accordance with some embodiments. The outer layer **210** covers the middle layer **190**, in accordance with some embodiments.

[0121] The materials and the structures of the inner layer **170**, the middle layer **190**, and the outer layer **210** are similar to or the same as that of the inner layer **170**, the middle layer **190**, and the outer layer **210** of FIGS. **4**A-**1** and **4**A-**2**, in accordance with some embodiments.

[0122] The inner spacer **630** is between the nanostructures **616** and between the nanostructures **601** and the source/drain structures **640**, in accordance with some embodiments. The inner spacer **630** is made of any suitable insulating material, such as silicon oxide, silicon oxynitride, borosilicate glass (BSG), phosphoric silicate glass (PSG), borophosphosilicate glass (BPSG), fluorinated silicate glass (FSG), low-k material, porous dielectric material, or a combination thereof.

[0123] The source/drain structures **640** are in direct contact with the fin portion **614** and the nanostructures **616**, in accordance with some embodiments. The source/drain structures **640** are positioned on two opposite sides of the gate stack G**3**, in accordance with some embodiments. The source/drain structures **640** include a source structure and a drain structure, in accordance with some embodiments.

[0124] The source/drain structures **640** are made of an N-type conductivity material, in accordance with some embodiments. The N-type conductivity material includes silicon phosphorus (SiP) or another suitable N-type conductivity material. The source/drain structures **640** are formed using an epitaxial process, in accordance with some embodiments.

[0125] The source/drain structures **640** are doped with the Group VA element, in accordance with some embodiments. The Group VA element includes phosphor (P), antimony (Sb), or another suitable Group VA material. In some embodiments, a concentration of the Group VA element (e.g. phosphor) ranges from about 3E21 atoms/cm^3 to about 7E21 atoms/cm^3. The source/drain structures **640** are also referred to as doped structures, in accordance with some embodiments.

[0126] In some other embodiments, the source/drain structures **640** are made of a P-type conductivity material, in accordance with some embodiments. The P-type conductivity material includes silicon germanium (SiGe) or another suitable P-type conductivity material. The source/drain structures **640** are formed using an epitaxial process, in accordance with some embodiments. The source/drain structures **640** are doped with the Group IIIA element, in accordance with some embodiments. The Group IIIA element includes boron or another suitable material.

[0127] Thereafter, as shown in FIGS. **7**B-**1** and **7**B-**2**, an upper portion of the outer layer **210** above the source/drain structures **640** is removed, in accordance with some embodiments. In some embodiments, a lower portion **212** of the outer layer **210** remains between the source/drain structures **640** and the gate stack G**3**. The lower portion **212** has a top surface **212***a*, in accordance with some embodiments.

[0128] As shown in FIGS. **7**B-**1** and **7**B-**2**, a recess **212***r* is formed in the top surface **212***a* of the lower portion **212**, in accordance with some embodiments. The top surface **212***a* is lower than top surfaces **642** of the source/drain structures **640**, in accordance with some embodiments. The removal process includes an isotropic etching process, such as a wet etching process, in accordance with some embodiments. In some embodiments, the outer layer **210** is made of silicon nitride, and the etchant used in the wet etching process includes phosphoric acid.

[0129] Afterwards, as shown in FIGS. **7**C-**1** and **7**C-**2**, an upper portion of the middle layer **190** above the source/drain structures **640** is removed, in accordance with some embodiments. As shown in FIG. **7**C-**2**, a lower portion **192** of the middle layer **190** remains between the outer layer **210** and the gate stack G**3**, in accordance with some embodiments.

[0130] The lower portion **192** has a top surface **192***a*, in accordance with some embodiments. The top surface **192***a* is lower than the top surface **642** of the source/drain structure **640**, in accordance with some embodiments. As shown in FIG. **7**C-**2**, a recess **192***r* is formed in the top surface **192***a* of the lower portion **192**, in accordance with some embodiments.

[0131] The recess **212***r* in the top surface **212***a* of the outer layer **210** and the recess **192***r* in the top surface **192***a* of the middle layer **190** together form a recess R**1**, in accordance with some embodiments. The recess R**1** has a W-like shape, in accordance with some embodiments. The depth D**1** of the recess **212***r* ranges from about 1 nm to about 6 nm, in accordance with some embodiments. The depth D**2** of the recess **192***r* ranges from about 1 nm to about 6 nm, in accordance with some embodiments.

[0132] The recess **212***r* has opposite edges **212**E**1** and **212**E**2**, in accordance with some embodiments. The edge **212**E**1** is adjacent to the source/drain structure **640**, in accordance with some embodiments. The edge **212**E**2** is adjacent to the middle layer **190**, in accordance with some embodiments. The edge **212**E**1** is higher than the edge **212**E**2**, in accordance with some embodiments.

[0133] The recess **192***r* has edges **192**E**1** and **192**E**2**, in accordance with some embodiments. The edge **192**E**1** is adjacent to the gate stack G**3**, in accordance with some embodiments. The edge **192**E**2** is adjacent to the outer layer **210**, in accordance with some embodiments. The edge **192**E**1** is higher than the edge **192**E**2**, in accordance with some embodiments.

[0134] The lower portion **201***a* of the spacer structure **201** is thicker than an upper portion **201***b* of the spacer structure **201**, in accordance with some embodiments. The thickness T**201***a* of the lower portion **201***a* is greater than the thickness T**201***b* of the upper portion **201***b*, in accordance with some embodiments. The spacer structure **201** has an L-like shape, in accordance with some embodiments.

[0135] Thereafter, as shown in FIGS. **7**D-**1** and **7**D-**2**, an etch stop layer **230** is formed over the source/drain structure **640**, the spacer structure **201**, and the mask layer M**2**, in accordance with some embodiments. The etch stop layer **230**

is filled into the recess **192**_r_ of the middle layer **190** and the recess **212**_r_ of the outer layer **210**, in accordance with some embodiments.

[0136] The etch stop layer **230** is made of an insulating material, such as a nitrogen-containing material (e.g., silicon nitride), in accordance with some embodiments. The etch stop layer **230** is formed using a deposition process, such as a chemical vapor deposition (CVD) process, an atomic layer deposition (ALD) process, or a physical vapor deposition (PVD) process, in accordance with some embodiments.

[0137] Afterwards, as shown in FIGS. **7**D-**1** and **7**D-**2**, a dielectric layer **240** is formed over the etch stop layer **230**, in accordance with some embodiments. The dielectric layer **240** is made of any suitable insulating material, such as silicon oxide, silicon oxynitride, borosilicate glass (BSG), phosphoric silicate glass (PSG), borophosphosilicate glass (BPSG), fluorinated silicate glass (FSG), low-k material, porous dielectric material, or a combination thereof. The dielectric layer **240** is deposited by any suitable process, such as a CVD process, a spin-on process, a sputtering process, or a combination thereof, in accordance with some embodiments.

[0138] Thereafter, as shown in FIGS. **7**E-**1** and **7**E-**2**, the steps of FIGS. **4**E-**1** to **4**H-**1** and **4**E-**2** to **4**H-**2** are performed to remove the mask layers M**1** and M**2**, the upper portions of the dielectric layer **240**, the etch stop layer **230**, and the spacer structure **201**, and the gate stack G**3** and to form a gate stack G**4**, the contact structures **280**, and the silicide layer SL, in accordance with some embodiments.

[0139] The gate stack G**4** is wrapped around the nanostructures **616** and an upper part of the fin portion **614**, in accordance with some embodiments. The gate stack G**4** includes a gate dielectric layer **250**, a work function metal layer **260**, and a gate electrode layer **270**, in accordance with some embodiments. The gate dielectric layer **250**, the work function metal layer **260**, and the gate electrode layer **270** are sequentially stacked over the nanostructures **616** and the upper portion of the fin portion **614**, in accordance with some embodiments.

[0140] The gate dielectric layer **250** is made of a dielectric material, such as a high dielectric constant (high-k) material, in accordance with some embodiments. The high-k material is made of metal oxides, such as hafnium oxide (HfO_2), hafnium silicon oxide (HfSiO), hafnium silicon oxynitride (HfSiON), hafnium tantalum oxide (HfTaO), hafnium titanium oxide (HfTiO), hafnium zirconium oxide (HfZrO), aluminum oxide, hafnium dioxide-alumina (HfO_2—Al_2O_3) alloy, other suitable high-k dielectric materials, or combinations thereof, in accordance with some embodiments. In some other embodiments, the high-k material is made of metal nitrides, metal silicates, transition metal-oxides, transition metal-nitrides, transition metal-silicates, oxynitrides of metals, other suitable materials, or combinations thereof.

[0141] The work function metal layer **260** provides a desired work function for transistors to enhance device performance including improved threshold voltage. In the embodiments of forming an NMOS transistor, the work function metal layer **260** can be an n-type metal capable of providing a work function value suitable for the device, such as equal to or less than about 4.5 eV. The n-type metal may be made of metal, metal carbide, metal nitride, or a combination thereof. For example, the n-type metal is made of tantalum, tantalum nitride, or a combination thereof.

[0142] In the embodiments of forming a PMOS transistor, the work function metal layer **260** can be a p-type metal capable of providing a work function value suitable for the device, such as equal to or greater than about 4.8 eV. The p-type metal may be made of metal, metal carbide, metal nitride, other suitable materials, or a combination thereof. For example, the p-type metal is made of titanium, titanium nitride, other suitable materials, or a combination thereof.

[0143] The gate electrode layer **270** is made of a suitable metal material, such as aluminum, tungsten, gold, platinum, cobalt, another suitable metal, an alloy thereof, or a combination thereof, in accordance with some embodiments.

[0144] The contact structures **280** are formed in the dielectric layer **240** and the etch stop layer **230**, in accordance with some embodiments. The silicide layer SL is formed between the contact structures **280** and the source/drain structures **640**, in accordance with some embodiments. In this step, a semiconductor device structure **600** is substantially formed, in accordance with some embodiments.

[0145] FIG. **8**A is a cross-sectional view of a semiconductor device structure **800**, in accordance with some embodiments. FIG. **8**B is a cross-sectional view of a semiconductor device structure **800**, in accordance with some embodiments.

[0146] As shown in FIGS. **8**A and **8**B, the semiconductor device structure **800** is similar to the semiconductor device structure **600** of FIGS. **7**E-**1** and **7**E-**2**, except that the middle layer **190** further has an upper portion **194**, in accordance with some embodiments. The upper portion **194** is thinner than the lower portion **192** of the middle layer **190**, in accordance with some embodiments.

[0147] Processes and materials for forming the semiconductor device structures **500**, **600**, and **800** may be similar to, or the same as, those for forming the semiconductor device structure **400** described above. Elements designated by the same or similar reference numbers as those in FIGS. **1** to **8**B have the same or similar structures and the materials. Therefore, the detailed descriptions thereof will not be repeated herein.

[0148] In accordance with some embodiments, semiconductor device structures and methods for forming the same are provided. The methods (for forming the semiconductor device structure) remove an upper portion of a spacer structure to enlarge the space for accommodating a contact structure. A lower portion of the spacer structure remains to separate a gate stack from source/drain structures.

[0149] In accordance with some embodiments, a method for forming a semiconductor device structure is provided. The method includes forming a gate stack over a substrate. The method includes forming a spacer structure over a sidewall of the gate stack. The spacer structure has an inner layer, a middle layer, and an outer layer, the inner layer covers the sidewall of the gate stack, and the middle layer is between the inner layer and the outer layer. The method includes forming a source/drain structure in and over the substrate, wherein a portion of the spacer structure is between the source/drain structure and the gate stack. The method includes partially removing the outer layer, wherein a first lower portion of the outer layer remains between the source/drain structure and the gate stack. The method includes partially removing the middle layer, wherein a second lower portion of the middle layer remains between the source/drain structure and the gate stack.

[0150] In accordance with some embodiments, a method for forming a semiconductor device structure is provided. The method includes forming a gate stack over a substrate. The method includes forming a spacer structure over a sidewall of the gate stack. The spacer structure has an inner layer, a middle layer, and an outer layer, the inner layer conformally covers the sidewall of the gate stack, the middle layer covers the inner layer, and the outer layer covers the middle layer. The method includes forming a source/drain structure in and over the substrate. The method includes removing a first upper portion of the outer layer above the source/drain structure. The method includes removing a second upper portion of the middle layer above the source/drain structure.

[0151] In accordance with some embodiments, a semiconductor device structure is provided. The semiconductor device structure includes a substrate. The semiconductor device structure includes a gate stack over the substrate. The semiconductor device structure includes a spacer structure over a sidewall of the gate stack. The spacer structure has an inner layer, a middle layer, and an outer layer, the inner layer covers the sidewall of the gate stack, and the middle layer is between the inner layer and the outer layer. The semiconductor device structure includes a source/drain structure in and over the substrate, wherein a portion of the spacer structure is between the source/drain structure and the gate stack, and a first top surface of the middle layer is lower than a second top surface of the source/drain structure.

[0152] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A semiconductor device structure, comprising:

a substrate;

a gate stack over the substrate;

a spacer structure over a sidewall of the gate stack, wherein the spacer structure has an inner layer, a middle layer, and an outer layer, the inner layer covers the sidewall of the gate stack, and the middle layer is between the inner layer and the outer layer; and

a source/drain structure in and over the substrate, wherein a portion of the spacer structure is between the source/drain structure and the gate stack, and a first top surface of the middle layer is lower than a second top surface of the source/drain structure.

2. The semiconductor device structure as claimed in claim **1**, wherein a third top surface of the outer layer is lower than the second top surface of the source/drain structure.

3. The semiconductor device structure as claimed in claim **1**, wherein the first top surface of the middle layer has a first recess.

4. The semiconductor device structure as claimed in claim **3**, wherein a third top surface of the outer layer has a second recess.

5. The semiconductor device structure as claimed in claim **4**, wherein the first recess of the first top surface of the middle layer and the second recess of the third top surface of the outer layer together form a recess having a W-like shape in a cross-sectional view of the outer layer and the middle layer.

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